PART - III

Chapter 9

STEAM GENERATORS

9.1 INTRODUCTION

Steam is mainly required for power generation, process heating and space heating purposes. The capacity of the boilers used for power generation is considerably large compared with other boilers.

Due to the requirement of high efficiency, the steam for power generation is produced at high pressures and in very large quantities. They are very large in size and are of individual design depending on the type of fuel used.

The boilers generating steam for process heating are generally smaller in size and generate steam at a much lower pressure. They are simpler in design and are repeatedly constructed to the same design. Though most of these boilers are used for heating purposes, some, like locomotive boilers are used for power generation also. In this chapter, some simple types of boilers will be described.

A steam generator popularly known as boiler is a closed vessel made of high quality steel in which steam is generated from water by the application of heat. The water receives heat from the hot gases through the heating surfaces of the boiler. The hot gases are formed by burning fuel, may be coal, oil or gas. Heating surface of the boiler is that part of the boiler which is exposed to hot gases on one side and water or steam on the other side. The steam which is collected over the water surface is taken from the boiler through superheater and then suitable pipes for driving engines or turbines or for some industrial heating purpose. A boiler consists of not only the steam generator but also a number of parts to help for the safe and efficient operation of the system as a whole. These parts are called mountings and accessories.

9.2 CLASSIFICATION OF BOILERS

1. According to the relative position of water and hot gases, the boilers are classified as fire tube or smoke tube boilers and water tube boilers.

In fire tube boilers, hot gases pass through tubes which are surrounded with water. There may be a single or double tubes as in the case of Lancashire boiler or there may be a bank of tubes as in a locomotive boiler. The Cochran, Lancashire and Locomotive are examples of this type of boilers. Due to their simplicity and because of the small capacity requirements of individual users, these boilers are universally used in many industries like paper, sugar and chemical industries for producing process steam.

In water tube boilers, the water circulates through the tubes and hot gases around them. Steam is generated inside the tubes and collected in a cylindrical vessel known as boiler drum. The boiler drum may not be exposed to the hot gases at all. This type of construction is flexible and so is found suitable for different capacity and pressure requirements, from process steam to power generation.

2. According to the axis of shell, the boilers are classified as vertical boilers and horizontal boilers. In vertical steam boilers, the axis of the shell is vertical whereas it is horizontal in case of horizontal steam boilers.

3. According to the method of the furnace, the boilers are classified as externally fired and internally fired boilers.

In externally fired boilers, the furnace is placed outside the boiler shell. The advantage of this type is that the fire place is simple and may be enlarged easily. Water tube boilers are always externally fired.

In internally fired boilers, the furnace is placed inside the boiler shell. Most of the fire tube boilers are of this type.

4. According to the method of water circulation, boilers are also classified as natural circulation and forced circulation boilers.

In natural circulation boilers, the water is circulated by natural convection currents which are set up due to the temperature difference. Most of the boilers of low capacity, fall under this category.

In forced circulation boilers, water is circulated with the help of a pump driven by motor. Forced circulation is used only in high pressure high capacity boilers.

5. According to the use of boiler, the boilers are classified as stationary and mobile boilers. Most of the industrial boilers and boilers used for power generation are of stationary type. The boilers used to run the locomotives and ships are mobile boilers as they continuously move from one place to other.

6. Boilers which are factory assembled and mounted on skids and are ready for operation, once water and steam lines are connected, are known as packaged boilers. These are generally of fire tube type of construction.

9.3 BOILERS' DETAILS

The details of the boilers are listed below :

1. Shell. The shell of the boilers is the main container usually of cylindrical shape, which contains water and steam.

2. Furnace. A furnace is another important part of the boiler. This may be a grate to burn coal or a burner to atomise and burn liquid fuel. Suitable area and volume should be provided for efficient combustion.

3. Water Flow Path. Water flow path is the path followed by the water in the boiler (particularly in water tube boiler) during the process of absorption of heat from hot gases and conversion into steam. The water should be free from dissolved material in order to reduce the scaling of the heating surface.

4. Gas Flow Path. The hot gas flow path either in fire tube or in water tube should be arranged in such a way that maximum heat of hot gases should be transferred to the water for steam generation. The boiler efficiency mainly depends upon the gas flow path.

5. Steam Path. In most of the boilers, the steam is taken out preferably at the top of the shell to avoid water particles being carried with the steam. To reduce the water particles carried by the steam, it is generally taken out through steam separators, in the case of large boilers.

6. Fittings. The valves and gauges which are necessary for the safety of the boiler, are known as mountings. Water level indicator, safety valve, blow-off cock and fusible plug are some of the mountings.

7. Accessories. Some equipments like economiser, air preheater and superheater are attached to the boiler to improve its overall efficiency. The economiser and air preheater are used to extract maximum heat from the flue gases and superheater is used to increase the temperature of steam above saturation temperature.

9.4 COCHRAN BOILER

Cochran boiler is a vertical, multi-tube boiler, commonly used for small capacity steam generation. It is made in different sizes of evaporative capacities ranging from 150 to 3000 kg per hour and working pressure upto 20 bar.

Figure 9.1 shows the arrangement of the boiler. The boiler consists of a cylindrical shell with its crown having a hemispherical shape. The furnace is also hemispherical in shape. The grate is placed at the bottom of the furnace and the ash-pit is located below the grate as shown in the figure. The coal is fed into the grate through the fire door and ash formed is collected in the ash-pit located jut below the grate and then it is removed manually. The furnace and the combustion chamber are connected through a pipe. The back of the combustion chamber is lined with fire bricks. The hot gases from the combustion chamber flow through the nest of horizontal fire tubes (generally 6.25 cm in external diameter and 165 to 170 in number). The gases passing through the fire tubes transfer a large portion of the heat to the water by convection. The flue gases coming out of fire tubes are finally discharged to the atmosphere through chimney.

The spherical top and spherical shape of fire-box are the special features of this boiler. These shapes require least material for the volume. The hemi spherical crown of the boiler shell gives maximum strength to withstand the pressure of the steam inside the boiler. The hemispherical crown of the fire box is advantageous for resisting intense heat. This shape is also advantageous for the absorption of the radiant heat from the furnace.

Coal or oil can be used as fuel in this boiler. If oil is used as fuel, no grate is provided but the bottom of the furnace is lined with fire bricks. Oil burners are fitted at a suitable location below the fire door.

A man-hole near the top of the crown of shell is provided for cleaning. In addition to this, a number of hand-holes are provided around the outer shell for cleaning purposes. The smoke box is provided with doors for cleaning of the interior of the fire tubes.

The air flow through the grate is caused by means of the draught produced by the chimney. A damper is placed inside the chimney (not shown) to control the discharge of hot gases from the chimney and thereby the supply of air to the grate is controlled. The chimney may also be provided with a steam nozzle (not shown) to discharge the flue gases faster through the chimney. The steam to the nozzle is supplied from the boiler.

The outstanding features of this boiler are listed below :

1. It is very compact and requires minimum floor area.
2. Any type of fuel can be used with this boiler.
3. It is well suited for small capacity requirements.
4. It gives about 70% thermal efficiency with coal firing and about 75% with oil firing.
5. The ratio of grate area to the heating surface area varies from 10 : 1 to 25 : 1.

It is provided with all required mountings. The function of each is briefly described below :

1. Pressure Gauge. This indicates the pressure of the steam in the boiler.

2. Water Level Indicator. This indicates the water level in the boiler. The water level in the boiler should not fall below a particular level otherwise the boiler will be overheated and the tubes may burn out.

3. Safety Valve. The function of the safety valve is to prevent the increase of steam pressure in the boiler above its design pressure. When the pressure increases above design pressure, the valve opens and discharges the steam to the atmosphere. When this pressure falls just below design pressure, the valve closes automatically. Usually the valve is spring controlled.

4. Fusible Plug. If the water level in the boiler falls below a predetermined level, the boiler shell and tubes will be overheated. And if it is continued, the tubes may burn as the water cover will be removed. It can be prevented by stopping the burning of fuel on the grate. When the temperature of the shell increases above a particular level, the fusible plug, which is mounted over the grate as shown in the figure, melts and forms an opening. The high pressure steam pushes the remaining water through this hole on the grate and the fire is *extinguished*.

5. Blow-off Cock. The water supplied to the boiler always contains impurities like mud, sand and salt. Due to heating, these are deposited at the bottom of the boiler, and if they are not removed, they are accumulated at the bottom of the boiler and reduces its capacity and heat transfer rates. Also the salt content will go on increasing due to evaporation of water. These deposited salts are removed with the help of blow-off cock. The blow-off cock is located at the bottom of the boiler as shown in the figure and is operated *only* when the boiler is running. When the blow-off cock is opened during the running of the boiler, the high pressure steam pushes the water and the collected material at the bottom is blown out. The concentration of salt is also reduced by blowing some water out. The blow-off cock is operated after every 5 to 6 hours of working for few minutes. This keeps the boiler clean.

6. Steam Stop Valve. It regulates the flow of steam supply outside. The steam from the boiler first enters into an antipriming pipe where most of the water particles associated with steam are removed.

7. Feed Check Valve. The high pressure feed water is supplied to the boiler through this valve. This valve opens towards the boiler only and feeds the water to the boiler. If the feed water pressure is less than the boiler steam pressure, then this valve remains closed and prevents the back flow of steam through the valve.

9.5 LANCASHIRE BOILER

It is stationary fire tube, internally fired, horizontal, natural circulation boiler. This is a widely used boiler because of its good steaming quality and its ability to burn coal of inferior quality. These boilers have a cylindrical shell 2 m in diameters and its length varies from 8 m to 10 m. It has two large internal flue tubes having diameter between 80 cm to 100 cm in which the grate is situated. This boiler is set in a brick work forming external flue so that the external part of the shell forms part of the heating surface.

The main features of the Lancashire boiler with its brick work shelling are shown in Fig. 9.2. The boiler consists of a cylindrical shell and two big furnace tubes pass right through this. One bottom flue and two side flues are formed by the brick setting. Both the flue tubes which carry hot gases lay below the water level as shown in the figure.

The grates are provided at the front end of the main flue tubes of the boiler and the coal is fed to the grates through the fire doors. A low fire brick bridge is provided at the end of the grate, as shown in the figure, to prevent the flow of coal and ash particles into the interior of the furnace tubes. Otherwise, the ash and coal particles carried with gases form deposits on the interior of the tubes and prevent the heat transfer to the water. The fire brick bridge also helps in deflecting the hot gases upward to provide better heat transfer.

The hot gases leaving the grate pass upto the back end of the tubes and then in the downward direction. They move through the bottom flue to the front of the boiler where they are divided into two and pass to the side flues as shown in the figure. Then they move along the two-side flues and come to the chimney as shown in the figure.

With the help of this arrangement of the flow passages of the gases, the bottom of the shell is first heated and then its sides. The heat is transferred to the water through surfaces of the two flue tubes (which remain in water) and bottom part and sides of the main shell. This arrangement increases the heating surface to a large extent.

Dampers in the form of sliding doors are placed at the end of side flues to control the flow of gases. This regulates the combustion rate as well as steam generation rate. These dampers are operated by chains passing over a pulley at the front of the boiler. This boiler is fitted with usual mountings. The pressure gauge and water level indicator are provided at the front whereas steam stop valve, safety valve, low-water and high steam safety valve and manhole are provided on the top of the shell.

The blow-off cock is situated beneath the front portion of the boiler shell for the removal of sediments and mud. It is also used to empty the water in the boiler whenever required for inspection.

The fusible plugs are mounted on the top of the main flues just over the grates as shown in the figure to prevent the overheating of boiler tubes by extinguishing the fire when the water level falls below a particular level. A low water level alarm is usually mounted in the boiler to give a warning in case the water level going below the precast value.

A feed check valve with a feed pipe is fitted on the front end plate. The feed pipe projecting into the boiler is perforated so that the water is uniformly distributed into the shell.

The outstanding features of this boiler are listed below :

1. Its heating surface area per unit volume of the boiler is considerably large.

2. Its maintenance is easy.

3. It is suitable where a large reserve of hot water is needed. Load fluctuations can be easily met by this boiler due to the large reserve capacity.

4. Superheater and economiser can be easily incorporated into the system, therefore, overall efficiency of the boiler can be considerably increased (80-85%).

The superheater is placed at the end of the main flue tubes. The hot gases before entering the bottom flue are passed over the superheater tubes as shown in the figure and the steam drawn through the team stop-valve is passed through the superheater. The steam passing through the superheater absorbs heat from hot gases and becomes superheated.

The economiser is placed at the end of side flues before exhausting the hot gases to the chimney. The water before being fed into the boiler through the feed check valve is passed through the economiser. The feed water is heated by absorbing the heat from the exhaust gases, thus leading to better boiler efficiency. Generally, a chimney is used to provide the draught.

9.6 LOCOMOTIVE BOILER

Locomotive boiler is a horizontal fire tube type mobile boiler. The main requirement of this boiler is that it should produce steam at a very high rate. Therefore, this boiler requires a large amount of heating surface and large grate area to burn coal at a rapid rate. The large heating surface area is provided by providing a large number of fire tubes and heat transfer rate is increased by creating strong draught by means of steam jet.

A modern locomotive boiler is shown in Fig. 9.3. It consists of a shell or barrel of 1.5 meter in diameter and 4 metres in length. The cylindrical shell is fitted to a rectangular fire box at one end and smoke box at the other end. The coal is manually fed on to the grates through the fire-door. The hot gases which are generated due to the burning of coal are deflected by a brick arch as shown in the figure. The fire box is entirely surrounded by narrow water spaces except for the fire hole and the ash-pit. The deflection of hot gases with the help of brick arch prevents the flow of ash and coal particles with the gases and it also helps for heating the walls of the fire-box properly and uniformly. It also helps in igniting the volatile matter from coal. The walls of the fire box work like an economiser. The ash-pit which is situated below the fire box is fitted with dampers at its front and back end shown in the figure to control the flow of air to the grate.

The hot gases from the fire box are passed through the fire tubes to the smoke box as shown in the figure. The gases coming to smoke box are discharged to the atmosphere through a short chimney with the help of a steam jet. All the fire tubes are fitted in the main shell. Some of these tubes (24 in number) are of larger diameter (13 cm diameter) fitted at the upper part of the shell and others (nearly 160 tubes) of 4.75 cm in diameter are fitted into the lower part of the shell. The shell contains water surrounding all the tubes. The top tubes are made of larger diameter to accommodate the superheater tubes. The steam passing through the superheater tubes is superheated by absorbing heat from the hot gases flowing over the tubes.

The steam generated in the shell is collected over the water surface. A dome-shaped chamber, known as steam dome, is fitted on the top of the shell. The dome helps to reduce the priming as the distance of the steam entering into the dome and water level is increased. The steam in the shell flows through a pipe mounted in the steam dome as shown in the figure into the steam header which is divided into two parts. One part of the steam header is known as saturated steam header and the other part is known as superheated steam header. The saturated wet steam through the steam pipe enters into the saturated steam header and then it is passed through the superheater tubes as shown in the figure. The superheated steam coming out of superheater tubes is collected in the superheated header and then fed to the steam engines. A stop valve serving also as regulator for steam flow is provided inside a cylindrical steam dome as shown in the figure. This is operated by the driver through a regulator shaft passing from the front of the boiler.

The supply of air to the grate is obtained by discharging the exhaust steam from the engine through a blast pipe which is placed below the chimney. The air-flow caused by this method is known as induced* draught. A large door at the front end of the smoke box is provided which can be opened for cleaning the smoke box and fire tubes.

The height of the chimney must be low to facilitate the locomotive to pass through tunnels and bridges. Because of the short chimney, artificial draught has to be created to drive out the hot gases. The draught is created with the help of exhaust steam when locomotive is moving and with the help of live steam when the locomotive is stationary. The motion of the locomotive helps not only to increase the draught, but also to increase the heat transfer rate.

The pressure gauge and water level indicators are located in the driver's cabin at the front of the fire box as shown in the figure. The spring loaded safety valve and fusible plug are

*In induced draught system, the flow of air is created by creating low pressure (below atmosphere) at the end of the gas path.

located as shown in the figure. Blow-of cock is provided at the bottom of the water wall to remove the debris and mud.

The outstanding features of this boiler are listed below :

1. Large rate of steam generation per square meter of heating surface. To some extent this is due to the vibration caused by the motion.

2. It is free from brickwork, special foundation and chimney. This reduces the cost of installation.

3. It is very compact.

The pressure of the steam is limited to about 20 bar.

The details of W.G. Type Locomotive boiler manufactured at Chittaranjan are listed below :

Diameter and length of shell	= 208.5 cm and 520.7 cm
Ordinary tubes	= 116 and 57.15 mm in diameter
Large size tubes	= 38 and 114.3 mm in diameter
Pressure and temperature of steam	= 14.76 bar and 370°C
Grate area	= 4.27 m^2
Heating surface area	= 270 m^2.

The capacity of this boiler under normal load is 8500 kg/hr at 14.76 bar and 370°C burning 158.5 kg of coal per hour/m^2 of grate area.

9.7 BABCOCK AND WILCOX BOILER

As classified earlier, in a water tube boiler, the water is inside the tubes and hot gases flow over the tubes. Babcock and Wilcox original model is a straight water tube boiler. A simple stationary boiler of this type is described here.

The boiler with its parts is shown in Fig. 9.4. The boiler shell known as water and steam drum is made of high quantity steel. It is connected by short tubes with the uptake header or riser and by longer tubes to the downtake header. The water level in the drum is slightly above the centre. The water tubes are connected to the top and bottom header and are kept inclined at an angle of 15° to the horizontal. The headers are provided with hand holes in the front of the tubes and are covered with caps. This arrangement helps in cleaning of the tubes. The inclined position helps the flow of water.

The furnace is arranged below the uptake header. Coal is fed to the grate through the fire door. Two firebrick baffles are arranged in such a manner that the hot gases from the grate are compelled to move in the upward and downward directions. First the hot gases rise upward and then go down and then rise up again and finally escape to the chimney through the smoke chamber.

The outer surface of the water tubes and half of the bottom cylindrical surface of the drum from the heating surface through which heat is transferred from the hot gases to the water.

The front portion of the water tubes comes in contact with the hot gases at higher temperature. So the water from this portion rises in the upper direction due to decreased density and passed into the drum through the uptake header. Here the steam and water are separated and the steam being lighter is collected in the upper part of the drum. From the back portion of the drum, the water enters into the water tubes through the downtake header. Thus, a continuous circulation of water from the drum to the water tubes and water tubes to the drum is maintained. The circulation of water is maintained by convective currents and is known as natural circulation.

A superheater is placed between the drum and water-tubes as shown in the figure. During the first turn of the hot gases, the gases are passed over the superheater tubes and the steam is passed through the superheater and becomes superheated steam. The steam is taken into the superheater from the steam space of the drum through a tube as shown in the figure. The superheated steam coming out through superheater is supplied through steam pipe and steam stop valve to the turbine. When the steam is being raised from cold boiler, the superheater is filled with water to the drum water level. This is essential to prevent the overheating of the superheater tubes. The superheater remains flooded with water until the steam reaches the working pressure. Once the rated pressure of steam is achieved in the boiler, then the water from the superheater is drained and steam is fed to it for superheating purposes.

A mud box is fitted to the down header as shown in the figure. The impurities and mud particles from the water are collected in the mud box and they are blown-off from time to time by means of a blow off valve as shown in the figure.

The access to the interior of the boiler is provided by the doors. This is necessary for cleaning the tubes and removing the soots from their surfaces. The draught is regulated by a damper which is provided in the back chamber as shown in the figure. The damper position is controlled with the help of chain connected to it from the pulley as shown in figure.

The outstanding features of this boiler are listed below :

1. The evaporative capacity of this boiler is high compared with other boilers (20,000 to 40,000 kg/hr). The operating pressure lies between 11.5 to 17.5 bar.

2. The draught loss is minimum compared with other boilers.

3. The defective tubes can be replaced easily.

4. The entire boiler rests over an iron structure, independent of brick work, so that the boiler may expand or contract freely. The brickwalls which form the surroundings of the boiler are only to enclose the furnace and the hot gases.

9.8 INDUSTRIAL BOILERS

The boilers are generally required in chemical industries, paper industries, pharmaceutical industries and many others. Efficiency, reliability and cost are major factors in the design of industrial boilers similar to central stations. Boiler's capacity varies from 100 to 400 tons of steam per hour. Industrial companies in foreign countries with large steam demands have considerable interest in cogeneration, the simultaneous production of steam and electricity because of federal legislation. High temperature and high pressure boilers (350°C and 75 bar) are now-a-days used even though high pressure and temperature are rarely needed to satisfy process requirement but they are used to generate electricity.

Due to surging prices of the oil, most of the industrial boilers are designed to use wood, municipal waste, pulverised coal, industrial solid waste and refinery gas.

Few industrial boilers which are in common use are discussed below :

1. Packaged Water-tube Boilers. The boilers having a capacity of 50 tons/hr are generally designed with water-cooled furnaces. Advantages of this design include minimum weight and maintenance as well as maximum structural rigidity and safety. A typical packaged boiler is shown in Fig. 9.5. Presently the packaged boilers are also designed to burn coal, wood and process waste also. The much larger furnace volumes required in units designed for solid fuels restrict the capacity of packaged units to about 40 tons/hr or about one-third of a oil-gas fired unit that can be shipped by railroad. Stoker fired systems are limited to 100 tons/hr capacity, while large boilers always burn pulverised coal. A boiler with a capacity of 50 tons/hr can be equipped with pulverised fuel when the price of coal rises because the pulverised firing provides 5% higher efficiency—an important consideration; specially in large capacity boilers.

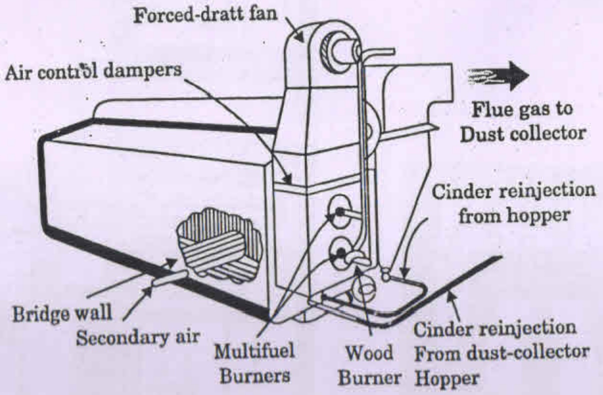
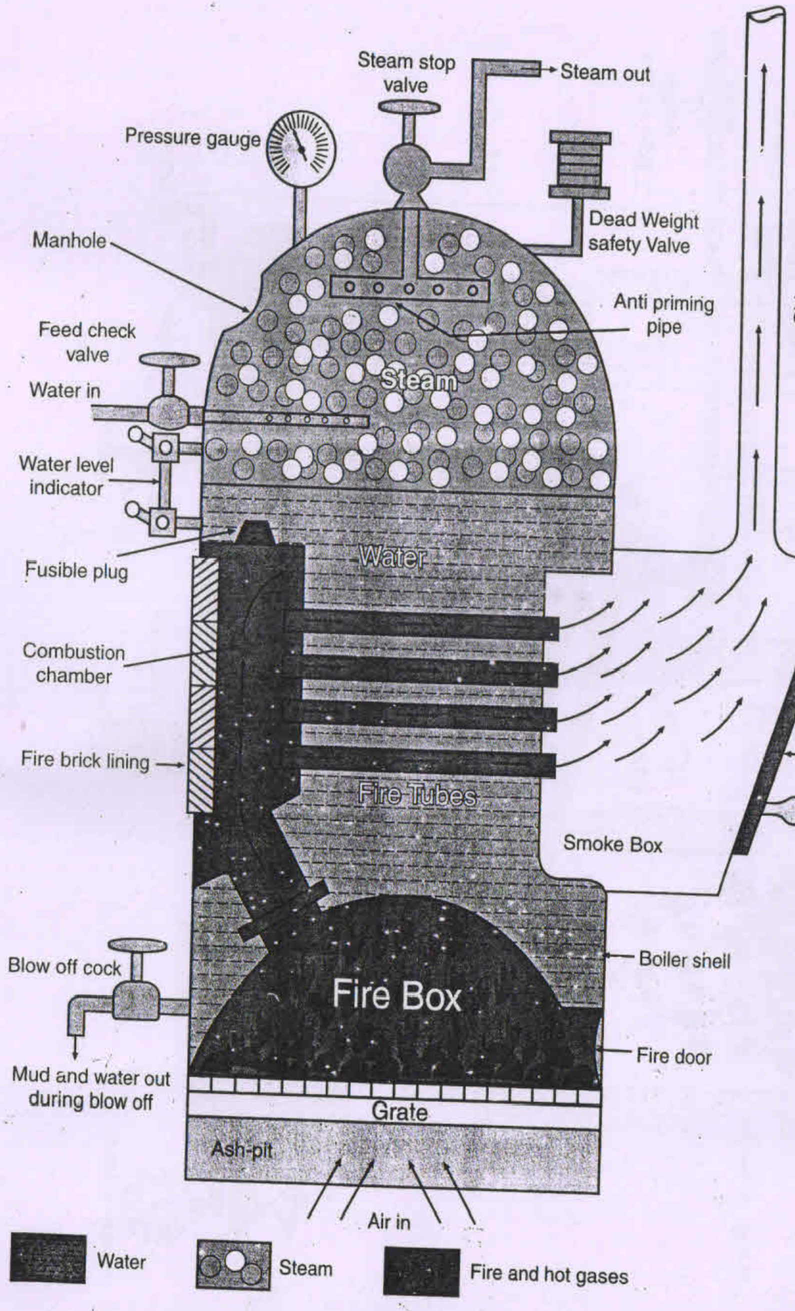


Fig. 9.5. Packaged boilers were extremely popular in the 1960s and early 1970s when oil and natural gas were cheap. They still are specifed—today in large numbers, but usually for service in small plants.

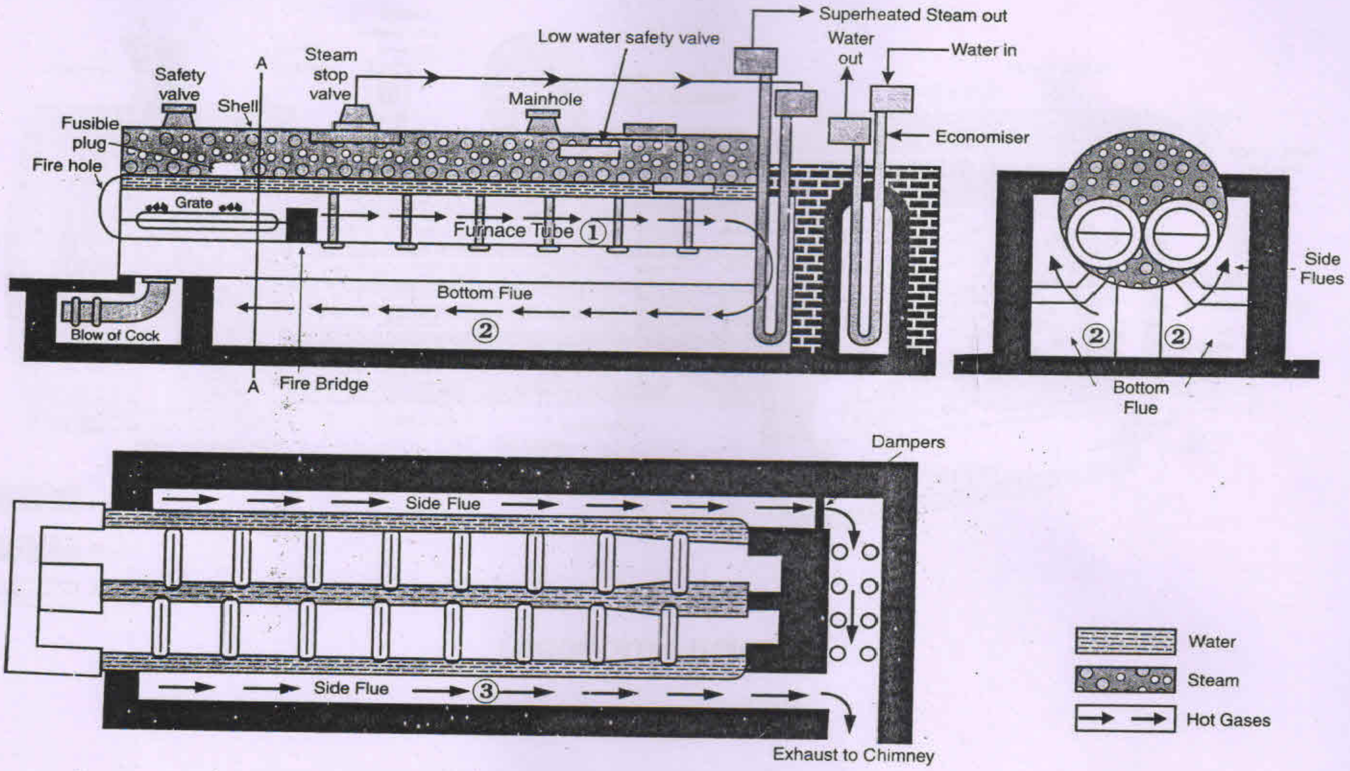
All field erected coal-fired boilers permit the combustion of variety of fuels, because of their conservative design.

2. Fluidized Bed Boilers (FBB). The coal based thermal power plants operating in India and abroad are of the pulverised fuel burning type requiring coal to be made into fine powder. Within 30-35% abrasive ash content in Indian coals, this process of coal preparation and subsequent erosion of boiler parts from ash are considered primary cause for power plant non-availability in the country. The solution to this problem is considered a fluidized bed boiler. In the fluidized bed boiler, coal upto 12 mm size can be burned while they are suspended in an agitated state within the combustor, using air blown in from the bottom. The process of heat transfer in such cases has been found to be extremely efficient, leading to more compact and economic boilers. The erosion problems are avoided in this method and wide range of other fuels like bagasse, rich husk, paper sludge and coal washery rejects can be used.

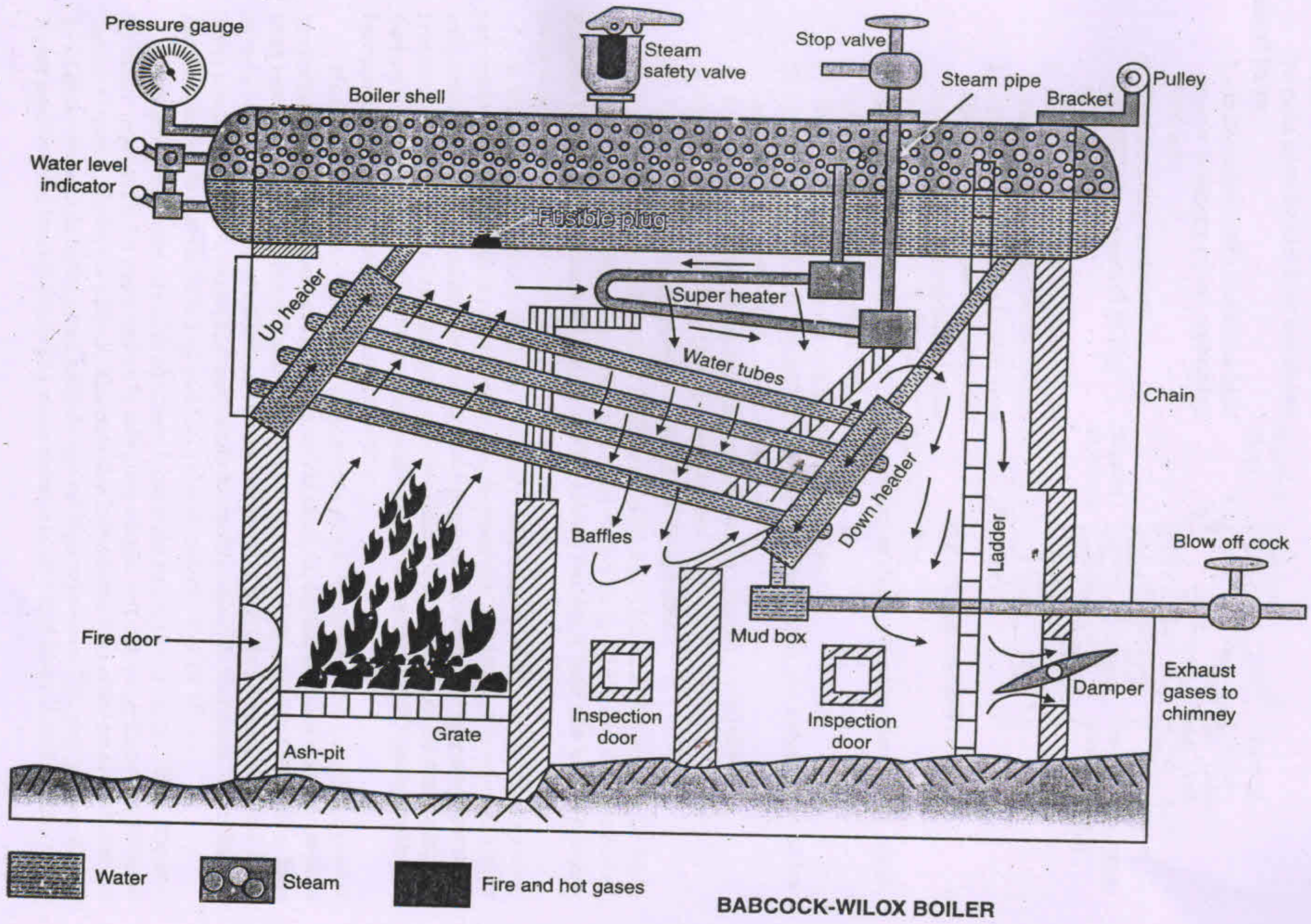
In the fluidized bed boiler, the solid fuel or waste material is in suspension with the help of air. The major problem with the coal fired boilers containing high sulphur is to suppress the SO_2 formed before exhausting the gases into the atmosphere as it is highly poisonous to the human health and crops. The FBB permits the injunction of lime stone directly into the furnace which can easily capture SO_2. This eliminates the need for expensive flue gas scrubbing system downstream of the boiler. The rating of this boiler goes even upto 150 tons/hr with a pressure of 150 bar and temperature of 400°C.



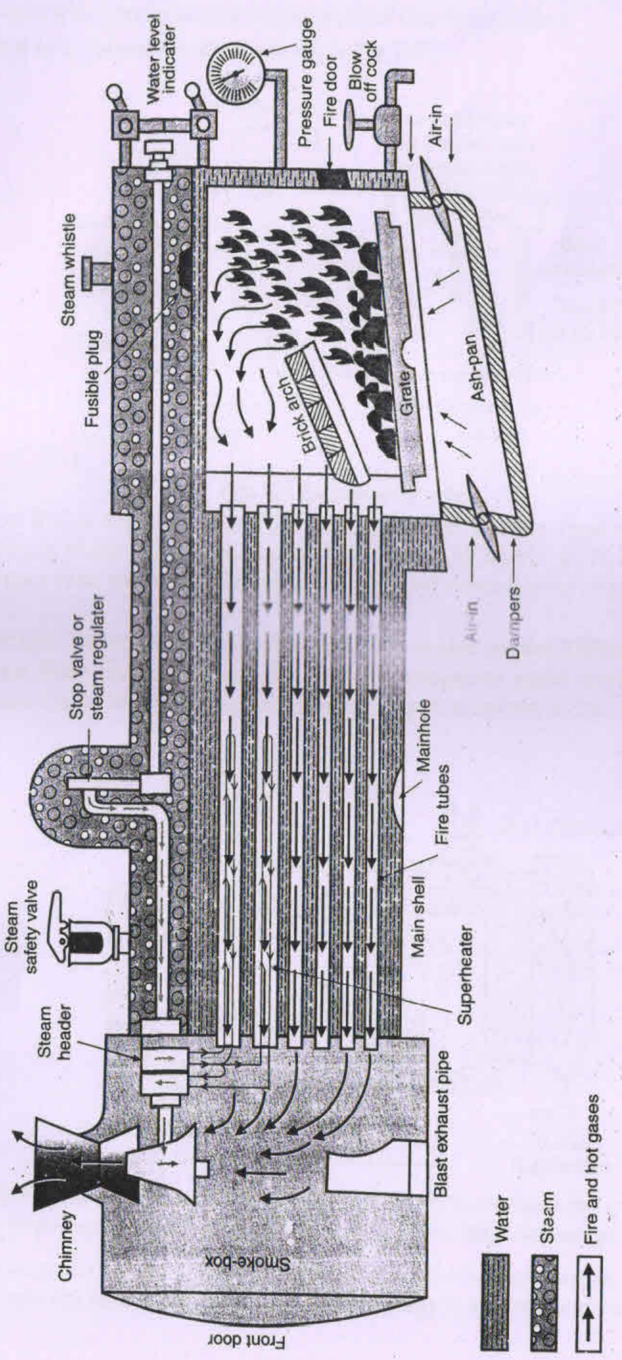
COCHRAN BOILER



LANCASHIRE-BOILER



BABCOCK-WILOX BOILER



LOCOMOTIVE-BOILER

A typical FBB boiler is shown in Fig. 9.6.

The main advantages of these boilers are listed below :

1. Ability to burn fuels containing high inerts (washery rejects containing 73% ash).
2. Fuel flexibility as fuel oil (40 MJ/kg) or rejects (8 MJ/kg) can be used.
3. Higher efficiency (75%) can be achieved even with inferior fuels.
4. As the bed temperature lies between 850 to 950°C, ash does not get heated to deformation temperature, hence the chances of grate clinkering or slagging are less.
5. Because of low temperature in combustion system, corrosion caused by alkali compounds in ash is significantly reduced.
6. The area required is much less than stoker system.
7. The emission of NO is reduced as chances of its formation are reduced due to low temperature in combustor.
8. SO_2 can be captured at the bed and its emission in exhaust is controlled.
9. Operation is as simple as that of oil-fired boiler.
10. Load response is comparable to that of an oil-fired boiler.

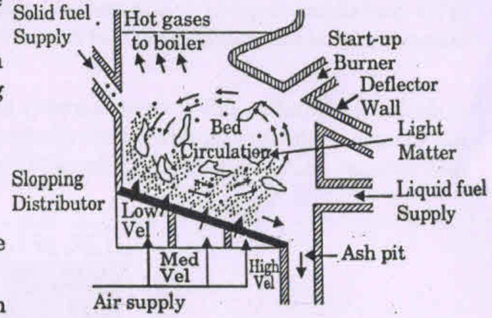


Fig. 9.6. Fluidized Bed Boiler.

A 12000 kg/hr, 14 bar and at 250°C FBC boiler at Trichry Chemicals designed by BHEL is in commercial operation since Oct. 1981. A team of engineers in BHEL has won the NRDC Republic Day Award for successful design of FBC to burn variety of fuels and high ash Indian Coals and washery rejects containing ash upto 73%.

3. Waste Heat Recovery Boilers. Recovery of heat exhausted from industrial processes and combustion equipment can often reduce fuel consumption and improve thermal efficiency with minimal capital investment. Depending on the properties of waste heat source temperature, pressure, flow rate, it may be attractive to generate steam or to use the heat for preheating air, feedwater or fuel oil. But when the temperature of waste stream exceeds 300°C, steam production becomes most economic method of heat recovery.

Many industries use the boilers to generate steam to recover the heat from process by-products and high temperature gas streams. For example, copper smelting industry requires heat recovery boilers capable of handling the hot, extremely dirty, sulphur laden gases from reverberatory furnaces, steel mills have boilers to burn dirty blast furnace gas, pulp and paper mills use black liquor to generate steam while recovering valuable chemicals and refineries have CO boilers to recover both heat and fuel value of catalytic cracker off-gas.

Designing conventional coal fired boilers to generate steam at 100 bar and 400°C required by many paper mills for cogeneration is relatively simple, but achieving these conditions in chemical recovery boilers is difficult. The corrosion in the lower part of the recovery boiler due to sulphur compounds in the gases militate against high pressures. The poor grades of black liquor need to blend troublesome liquid waste streams in with that by-product fuel, limit design flexibility. Thus material selection and furnace design are important considerations in recovery boilers when pressure and temperature of steam are higher.

A typical heat recovery boiler is shown in Fig. 9.7.

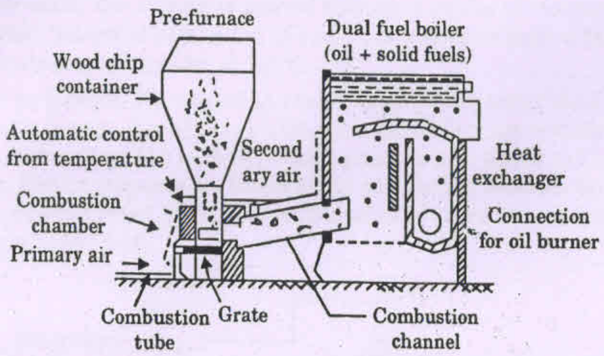


Fig. 9.7. Heat recovery boiler.

4. Fire Tube Boilers. The fire tube boilers of firebox type are used in many industrial plants where the steam demand is less than 25 tons/hr at 15 bar pressure. They are also cheaper than water tube boilers in this range of capacity when liquid or gaseous fuels are used.

In firetube boilers, the fire tubes heat the water in the boiler while conveying burned gases from the furnace to the stack. Combustion gases normally make these passes through the boiler before being discharged. A typical industrial fire tube boiler is shown in Fig. 9.8.

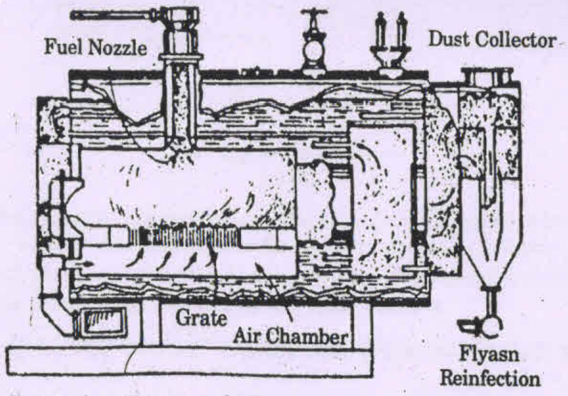


Fig. 9.8. Fire tube boiler capable of burning solid fuels in institutional and small industrial plants has stationary grate. Dropping tube shown distributes coal on grate.

The advantages of fire tube over water tube boilers include easier tube replacement, minimum headroom requirements and low susceptibility to cold end and casing corrosion.

5. Waste Heat Boiler with Dual Combustion Chamber. Solid wastes are most commonly burned in dual chamber as shown in Fig. 9.9. A ram type feeder injects the waste into primary combustion chamber where the material is reduced to inert ash weighing approximately 5% of the initial charge. Entrained particulates and gases pass into the secondary combustion chamber where they are burned.

Either batch or continuous waste-feeding system is used in this boiler. Residual ash is removed manually from batch units and automatically from continuous models. Automatic ash removal system consists of a conveyor that moves the ash along the combustion chamber floor and discharges it from the unit.

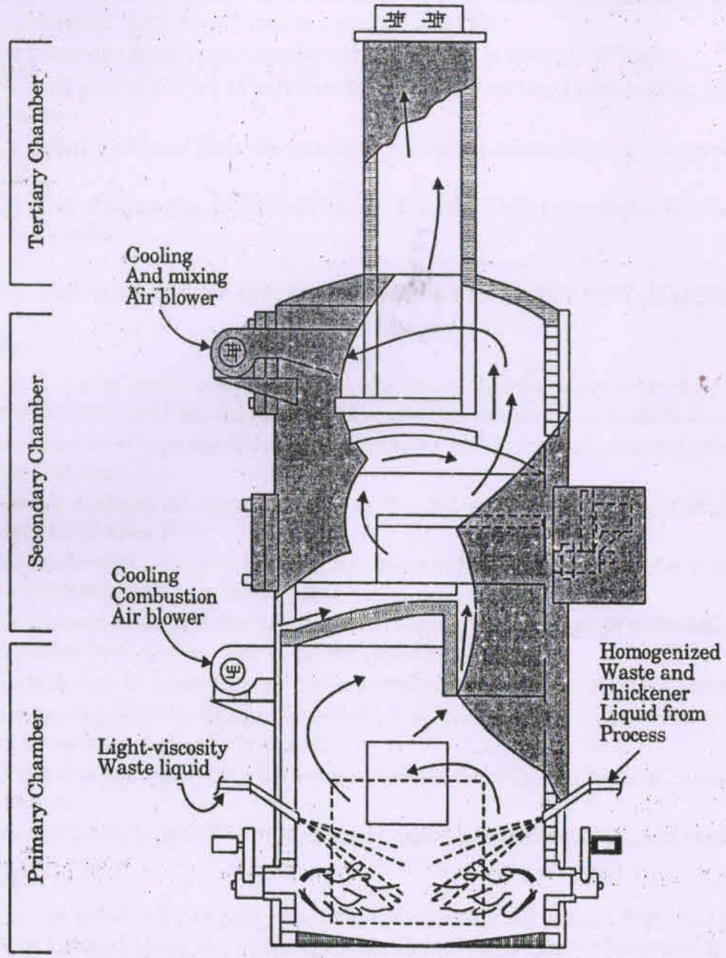


Fig. 9.9. Liquid-waste and sludge incinerator uses atomizing nozzles to inject waste into primary combustion chamber.

6. Electric Boilers. The use of electric boilers in lower capacity range—has become popular due to high prices of the oil and pressure put by the Govt. to clean up the exhaust gases. There are two basic types of electric boilers—resistance and electrode.

In resistance units, the current is passed through a series of resistances which are submerged in water. Industrial application of resistance boilers is limited because they are economical only in sizes upto 5 tons/hr. at 100°C.

In electrode type boiler, the current is passed through the water itself and the water converts the electrical energy directly into steam. Electrode boilers are available for either low voltage (600 V) or high voltage (2.3 to 15 kV). Steam pressure upto 30 bar and 200°C is possible with these boilers. Higher pressures are theoretically possible but insulator breakdown occurs above 250°C and pressure vessel costs increase dramatically above 20 bar. A typical electrode type boiler is shown in Fig. 9.10.

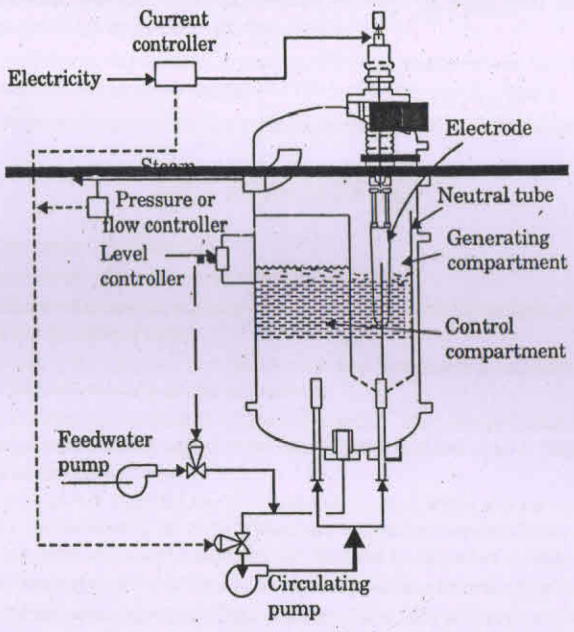


Fig. 9.10. High-voltage boiler produces steam as current flows from electrodes to the neutral walls of the cylinders around them.

Note : For high pressure boilers used in thermal power plants, the students are advised to see the book **Power Plant Engineering** by the same authors.

9.9 MERITS AND DEMERITS OF FIRE TUBE BOILERS OVER WATER TUBE BOILERS

Merits :

1. The fire tube boilers have greater reliability and low first cost because of simple and rigid construction.
2. Due to large cylindrical drums of fire tube boilers, there is ample water surface from which the system can be quickly raised. Simple antipriming devices serve the purpose very well.

3. Fire tube boilers are excellent for engines operating with rapid changes in load like locomotive boiler.
4. Fire tube boilers are very compact and space occupied per kg of steam generation by fire tube boilers is considerably less than the water tube boilers.
5. Failure in feed water supply for some time does not cause damage to boiler as it contains large quantity of water.

Demerits :

1. Fire tube boilers have large ratio of water to steam and therefore makes the boiler slow in reaching the operating pressure.
2. Larger diameter of shell 2.4 m and limit of maximum thickness 3 cm and stress consideration limit the generation pressure to 20 bar.
3. The maximum generating capacity of these boilers is about 9000 kg/hr.
4. The explosion of the fire tube boilers becomes very serious because of its large water capacity.
5. The transportation of fire tube boilers is very inconvenient because of large size of the shell.
6. Slow rate of steam generation renders the fire tube boilers unsuitable for use in steam power plants.

9.10 MERITS AND DEMERITS OF WATER TUBE BOILERS OVER FIRE TUBE BOILERS

Merits :

1. Generation of steam is much quicker due to small ratio of water content to steam content. This also helps in reaching the steaming temperature in short time.
2. Its evaporative capacity is considerably larger and the steam pressure range is also high-200 bar.
3. Heating surfaces are more effective as the hot gases travel at right angles to the direction of water flow.
4. The combustion efficiency is higher because complete combustion of fuel is possible as the combustion space is much larger.
5. The thermal stresses in the boiler parts are less as different parts of the boiler remain at uniform temperature due to quick circulation of water.
6. The boiler can be easily transported and erected as its different parts can be separated.
7. Damage due to the bursting of water tube is less serious. Therefore, water tube boilers are sometimes called safety boilers.
8. All parts of the water tube boilers are easily accessible for cleaning, inspecting and repairing.
9. The water tube boiler's furnace area can be easily altered to meet the fuel requirements.

Demerits :

1. It is less suitable for impure and sedimentary water, as a small deposit of scale may cause the overheating and bursting of tube. Therefore, use of pure feed water is essential.
2. They require careful attention. The maintenance costs are higher.
3. Failure in feed water supply even for short period is liable to make the boiler overheated.

9.11 REQUIREMENTS OF A GOOD BOILER

A good boiler must possess the following qualities :

1. The boiler should be capable to generate steam at the required pressure and quantity as quickly as possible with minimum fuel consumption.
2. The initial cost, installation cost and the maintenance cost should be as low as possible.
3. The boiler should be light in weight, and should occupy small floor area.
4. The boiler must be able to meet the fluctuating demands without pressure fluctuations.
5. All the parts of the boiler should be easily approachable for cleaning and inspection.
6. The boiler should have a minimum of joints to avoid leaks which may occur due to expansion and contraction.
7. The boiler should be erected at site within a reasonable time and with minimum labour.
8. The water and flue gas velocities should be high for high heat transfer rates with minimum pressure drop through the system.
9. There should be no deposition of mud and foreign materials on the inside surface and soot deposition on the outer surface of the heat transferring parts.
10. The boiler should conform to the safety regulations as laid down in the *Boiler Act*.

EXERCISES

9.1. State how the boilers are classified.

9.2. Explain the principle of fire tube and water tube boilers.

9.3. Describe with a neat sketch the working of Cochran boiler. Show the position of different mountings and explain the function of each.

9.4. Describe, giving neat sketches, the construction and working of a Lancashire boiler. Show the positions of different mountings and accessories.

9.5. Sketch and describe the working of a Locomotive boiler. Show the positions of fusible plug, blow off cock, feed check valve and superheater. Mention the function of each. Describe the method of obtaining draught in this boiler.

9.6. Give an outline sketch showing the arrangement of water tubes and furnace of a Babcock and Wilcox boiler. Indicate on it, the path of the flue gases and water circulation. Show the positions of fusible plug, blow off cock and superheater. Mention the function of each.

9.7. Explain why the superheater tubes are flooded with water at the starting of the boilers ?

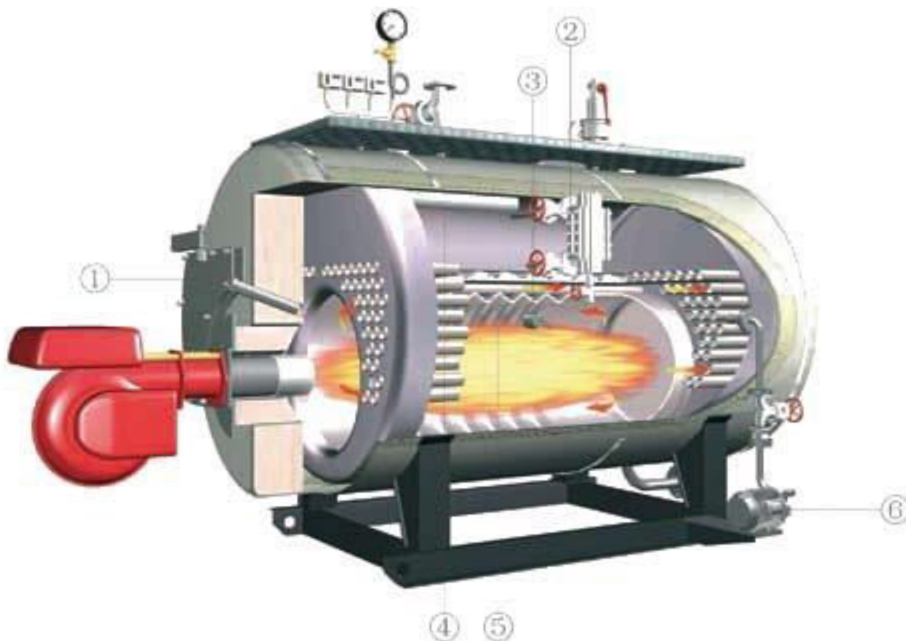
9.8. Mention the chief advantages and disadvantages of fire tube boilers over water tube boilers.

9.9. Discuss the chief advantages of water tube boilers over fire tube boilers.

9.10. What are the considerations which would guide you in selecting the type of boiler to be adopted for a specific purpose ?

9.11. Distinguish between water-tube and fire-tube boilers and state under what circumstances each type would be desirable.

BOILER MOUNTINGS



Engr. Omar Sadath, Maritme Lecturer
& Trainer, Bangladesh

overview

Mountings are crucial without which the operation of boilers is unsafe.

Proper maintenance and care of the mountings is important for the safety of both the boiler and the personnel and to maintain optimum operating condition of the boiler.

Boiler Mountings:

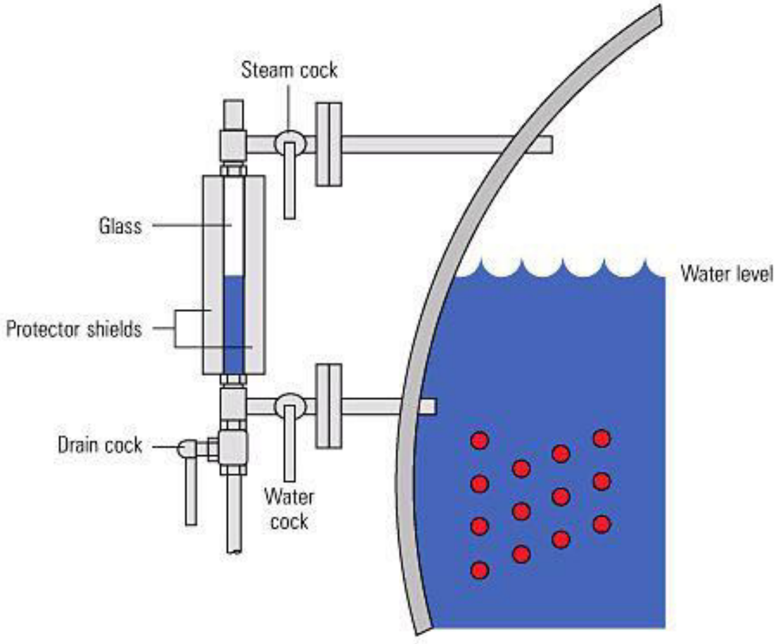
The boiler mountings are the part of the boiler and are required for proper functioning. In accordance with the Indian Boiler regulations, of the boiler mountings is essential fitting for safe working of a boiler.

List of mountings

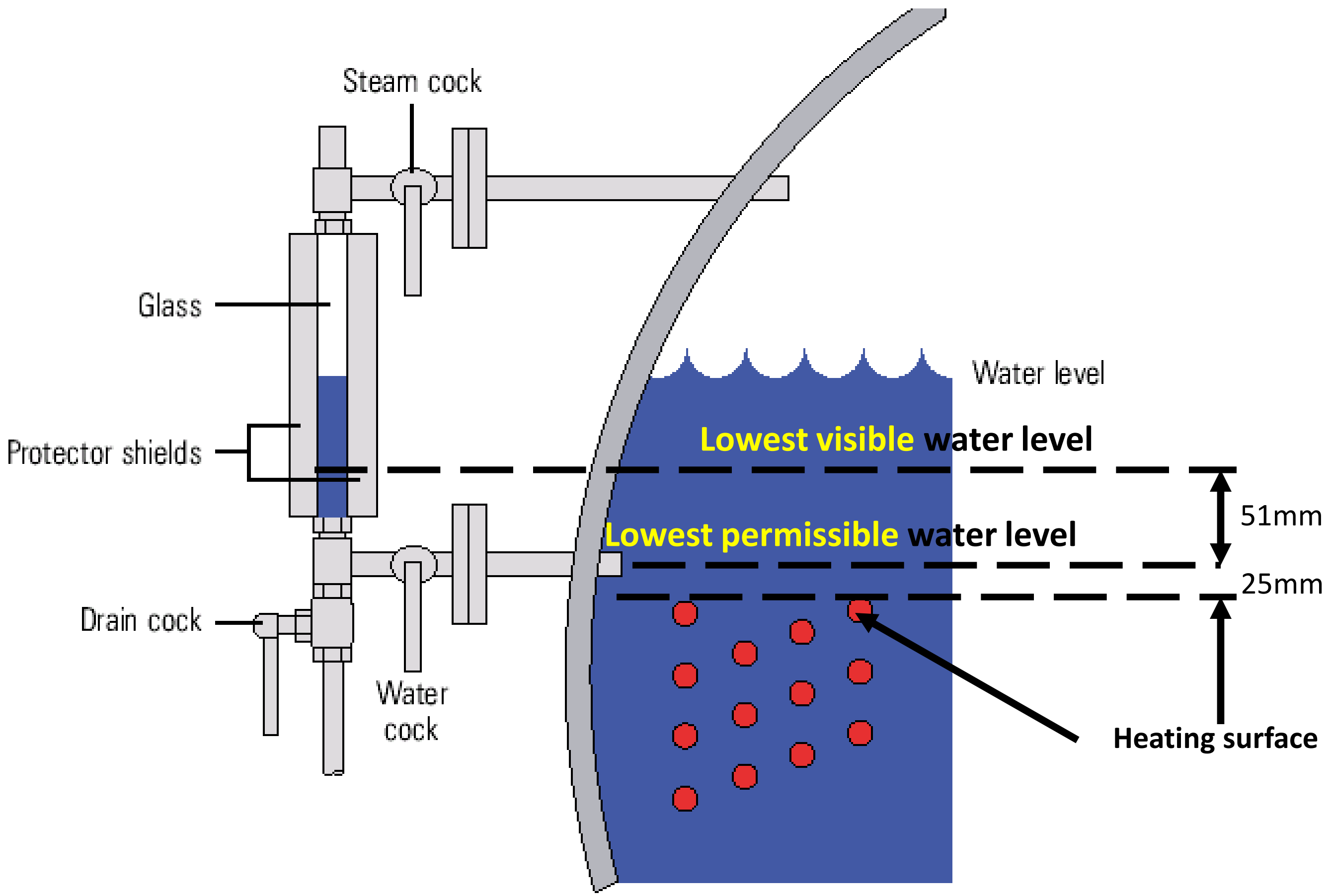
- Safety valves
- Water level indicators
- Water level controller
- Water level alarms & cut-out assembly
- Remote water level transmitter
- Main steam outlet valve
- Pressure gauge with cock & Pressure switches
- Feed water valves
- Burner assembly
- Air vent
- Water sampling valve
- Manholes, mud holes & peepholes
- Bottom blow down valve
- De-foaming (scum) valve
- Furnace drain valve
- Soot blowers

Water level Indicator

Water level indicator is located in front of boiler in such a position that the level of water can easily be seen by attendant. Two water level indicators are used on all boilers.



Gauge glass

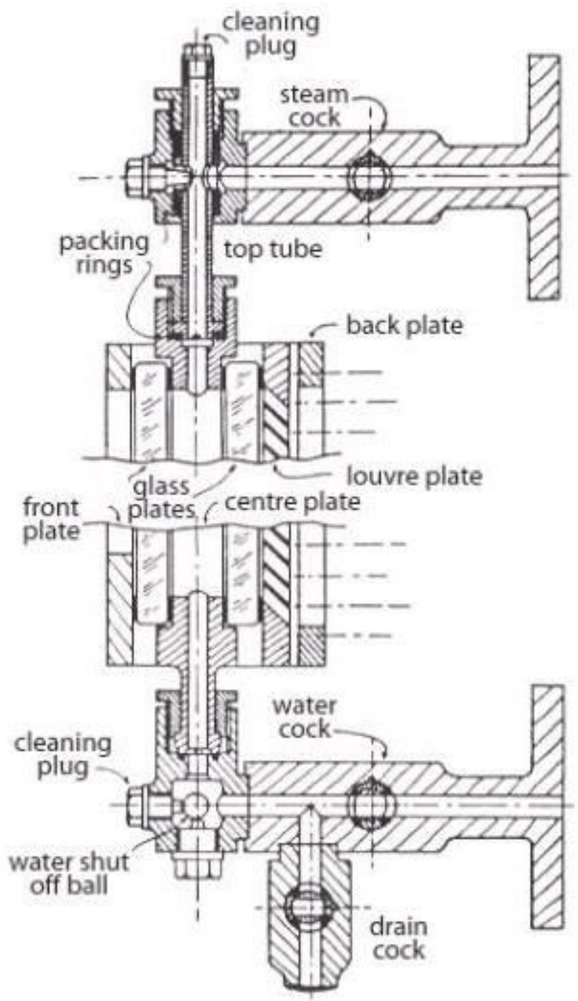


Steam cock
Glass
Protector shields
Drain cock
Water cock
Water level
Lowest visible water level
Lowest permissible water level
51mm
25mm
Heating surface

Type of level Indicator and Switches

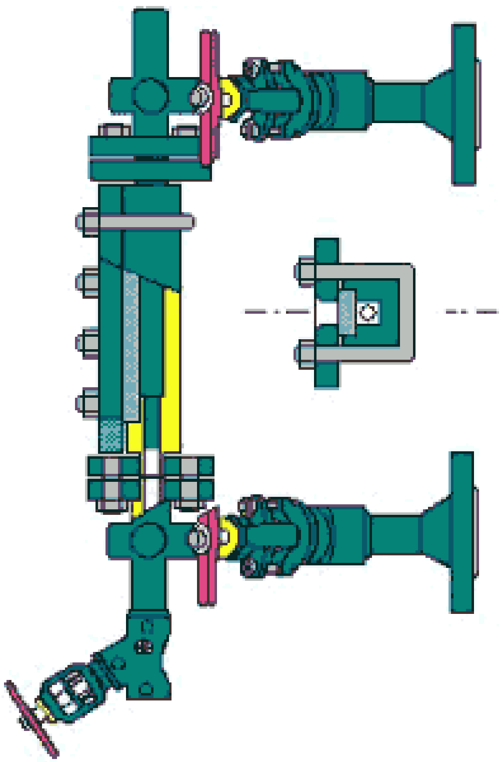
- Tubular gauge glass. Suitable for low pressure boiler of design pressure below 17 bar.
- Reflex plate gauge glass. Suitable for boilers design pressure below 34 bar.
- Double plate gauge glass. Suitable for boiler design above 34 bar.

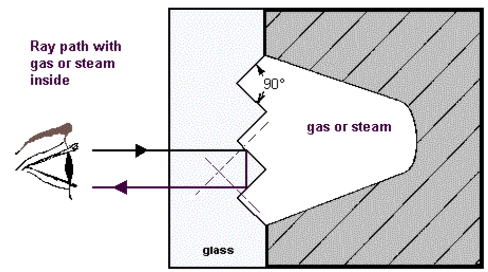
Gauge glass

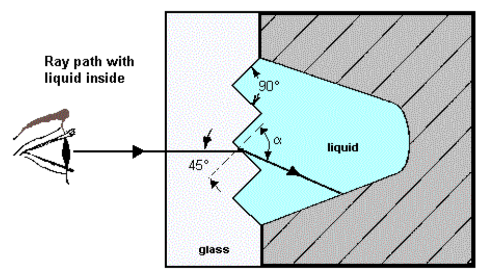


Double plate Gauge Glass

Gauge glass







Reflex plate Gauge
Glass

Regulation for Gauge Glass

- Every boiler is to be fitted with at least two independent means of water level indicator. The other means is to be either an additional glass gauge or an approved equivalent device.
- Water tube boilers are to be fitted two system of water level detection, which are to be independent of any other mounting on the boiler. Both the systems are to be operate audible and visible alarm and automatic shut off.

Regulation for Gauge Glass

- Gauge glass to be so located that the lowest visible water level in the glass is not lower than 51 mm(2 in) above the lowest permissible water level.
- The lowest permissible water level is just above (usually 25 mm or 1 inch above) the top row of tubes when cold.

How to repair a Gauge Glass

- Isolate the assembly by shutting of the steam and water connection and by opening the drain.
- Take off the screws of the glass holder and remove the pressure ledge.
- Remove the glass insert with gasket.
- Spare glass surface to be clean thoroughly.
- Install the black gasket ,the spare glass groove facing inwards,the red gasket ,finally the thin steel sheet.

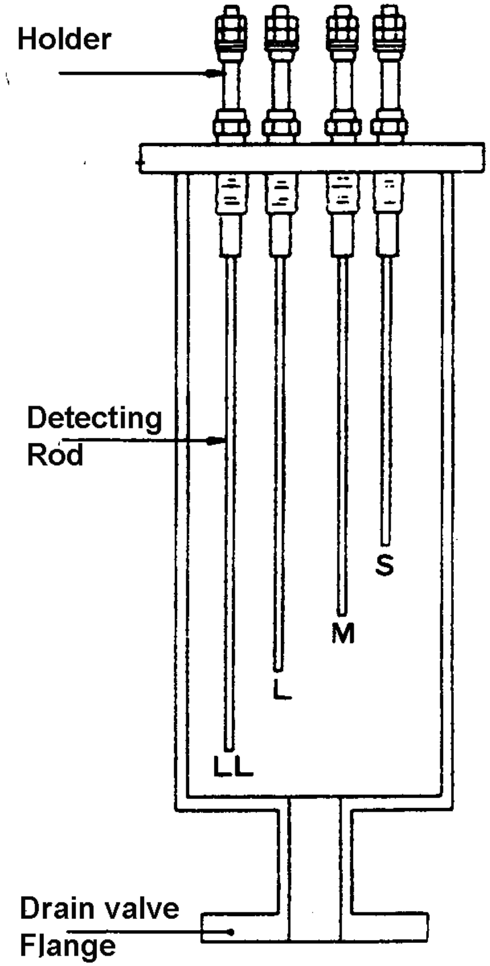
How to repair a Gauge Glass

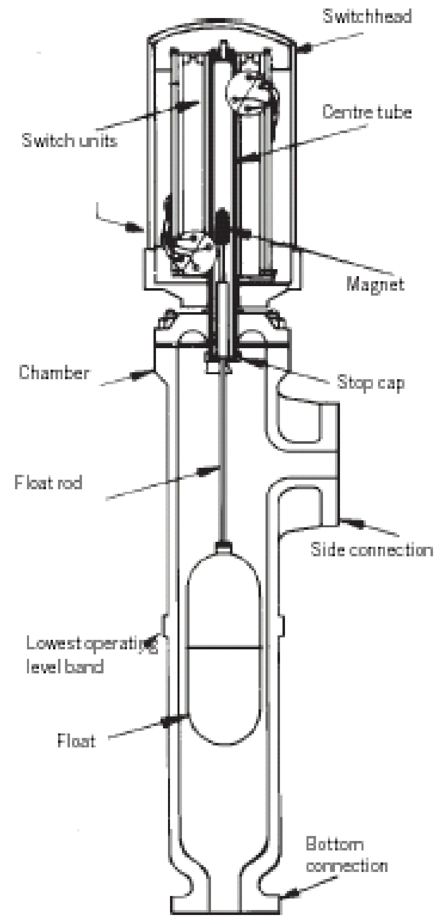
- Put the pressure ledges and tightened the screw uniformly starting from the middle and proceding crosswise up and downwards.
- Heat the new glass slowly by keeping the water side valve shut, the steam side valve crack open keeping the drain valve open.
- After 30 mins the screw should be retightened.
- The assembly can be put on load by shutting of the drain valve and opening the steam and water side valves completely open.

High low level alarm and burner cut-off

- All types of marine boiler must be fitted with high low water level safeguards. All such of equipment is to be capable of operating audible and visible alarms and also of automatic shutting off the fuel supply to the burners when the water level falls to predetermined low level.
- Two types of level alarms are commonly found.
- Magnetic float switch type, Conductivity probes type.

Level switch and Alarms

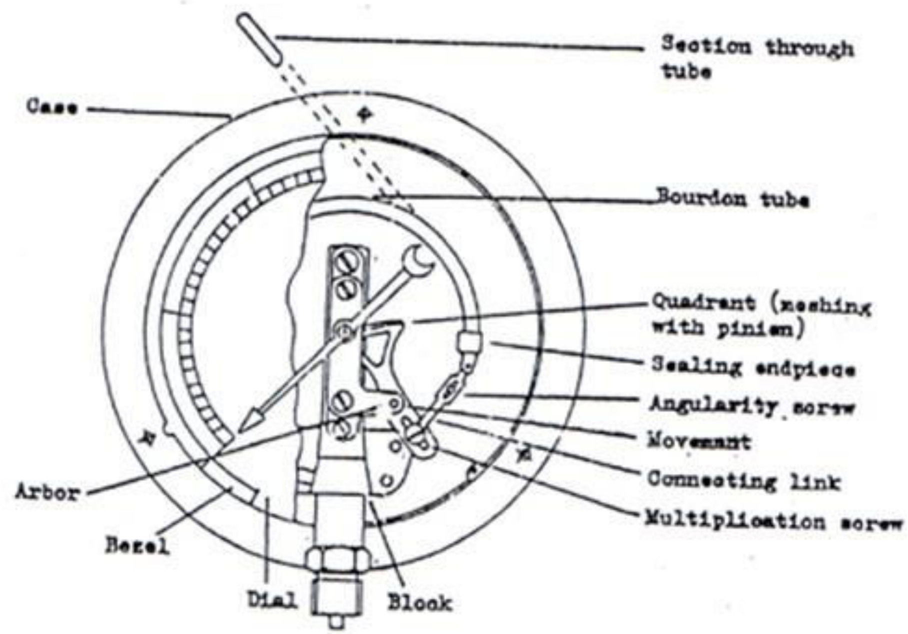




Low water level cutout failure

Pressure Gauge

A pressure gauge is fitted in front of boiler in such a position that the operator can
conveniently read it. It reads the pressure of steam in the boiler and is connected to steam space by a siphon tube.



The most commonly, the Bourdon pressure gauge is used.

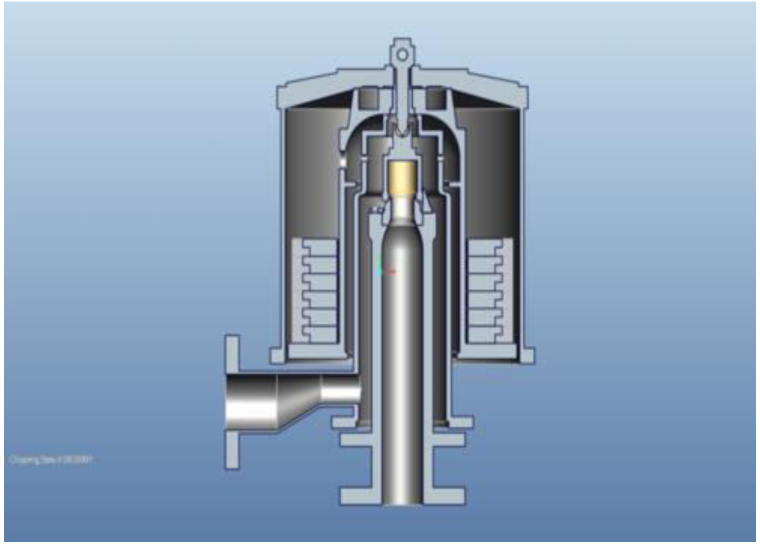
Safety Valve

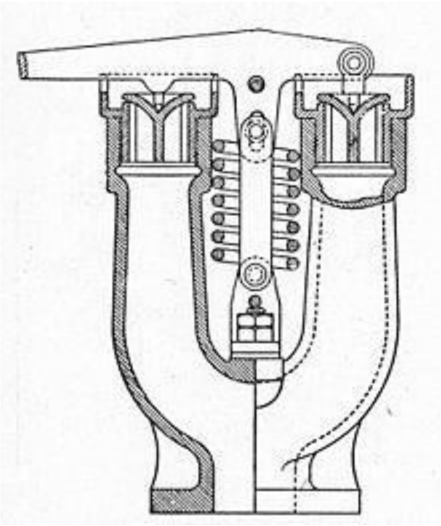
Safety valves are located on the top of the boiler. They guard the boiler against the excessive high pressure of steam inside the drum. If the pressure of steam in the boiler drum exceeds the working pressure then the safety valve allows blow-off the excess quantity of steam to atmosphere. Thus the pressure of steam in the drum falls. The escape of steam makes a audible noise to warn the boiler attendant.

There are four types of safety valves:

1. Dead weight safety valve.
2. Spring loaded safety valve
3. Lever loaded safety valve
4. High steam and low water safety valve

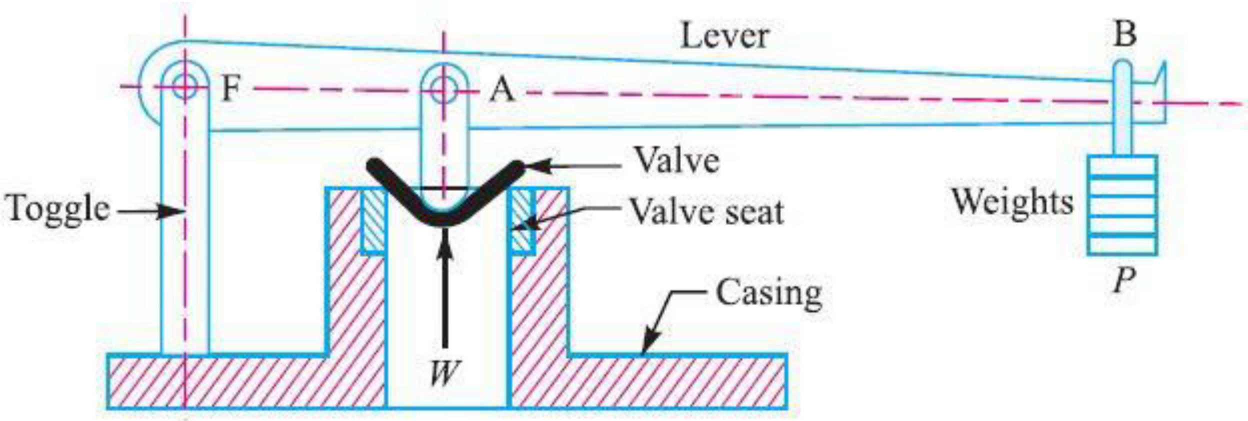
According to design feature safety valve can be classified as Ordinary lift, High lift, Improved high lift and full bore.





Dead weight safety valve

Spring loaded safety valve

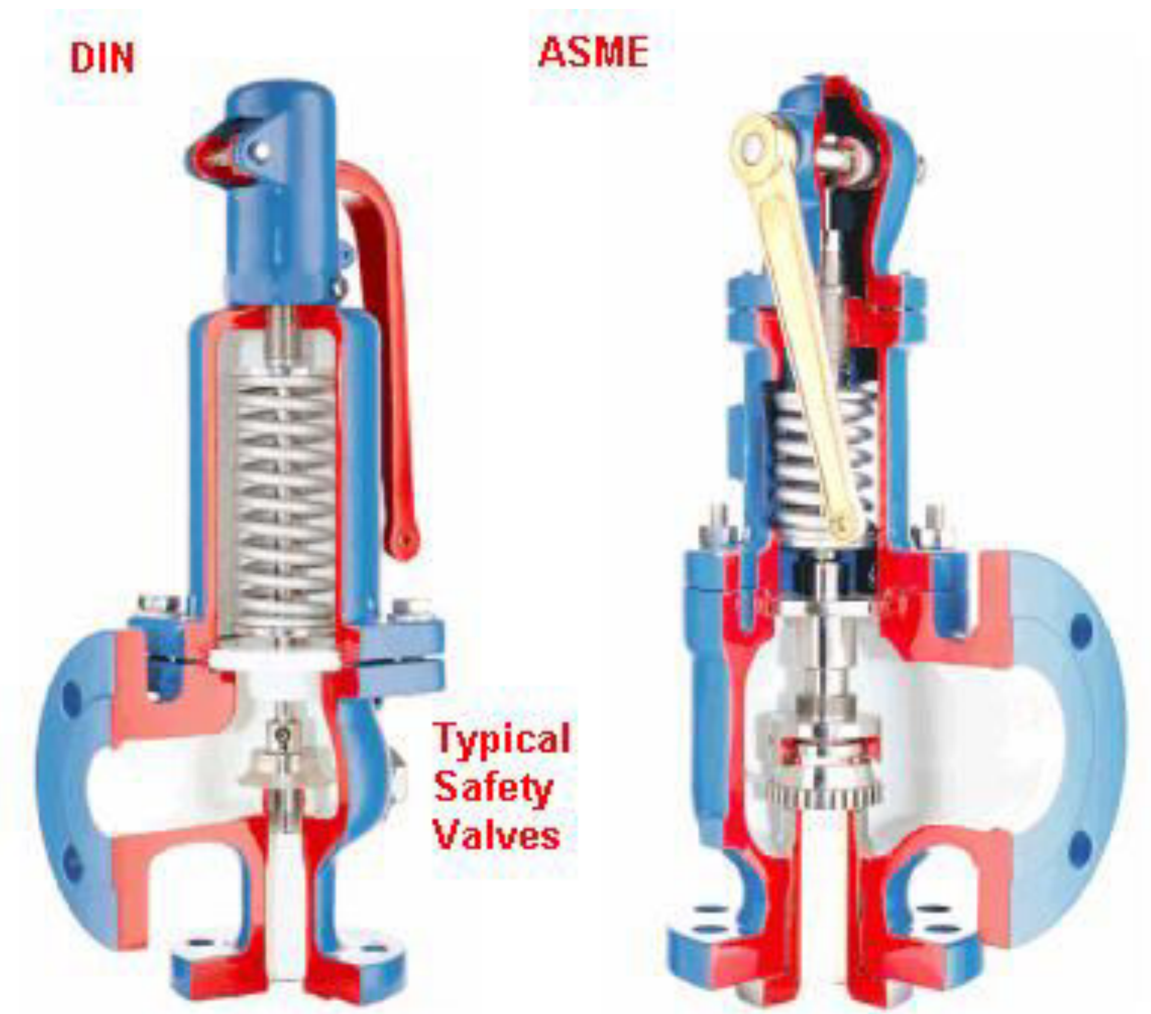


Lever loaded safety valve

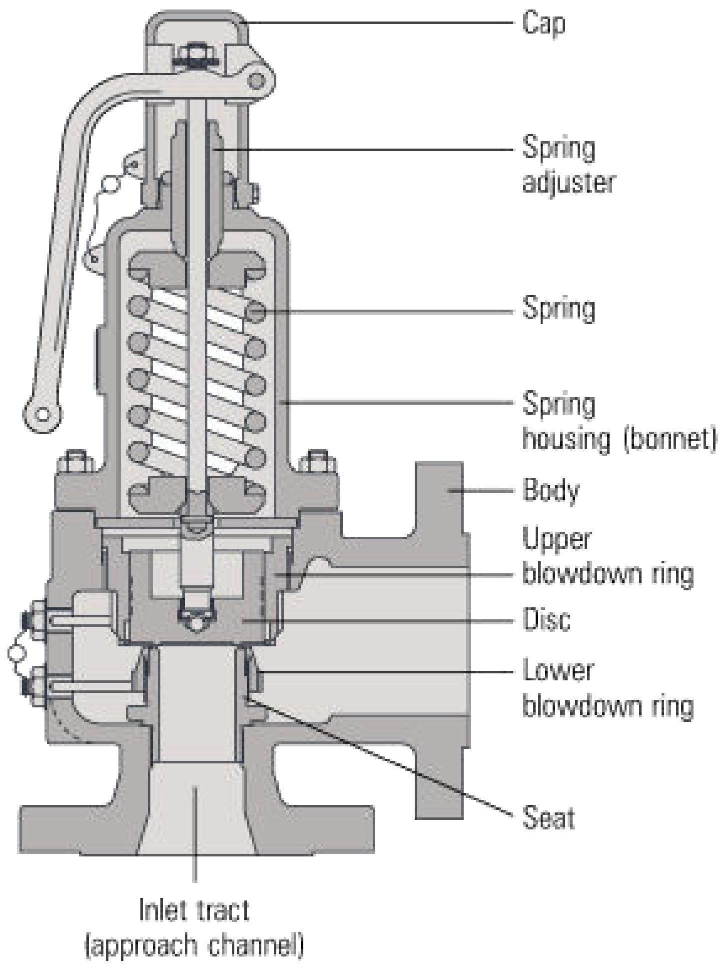
Safety valves- Definitions

- **Set Pressure:** The boiler pressure at which the safety valve begins to lift.
- **Closing Pressure:** The boiler pressure at which the valve closes.
- **Valve lift:**The axial valve disc travel from closed to the open position.
- **Blow down:** The difference between the opening and the closing pressures. Too much blow down causes wastage of steam, where as too little blow down would give unstable condition.

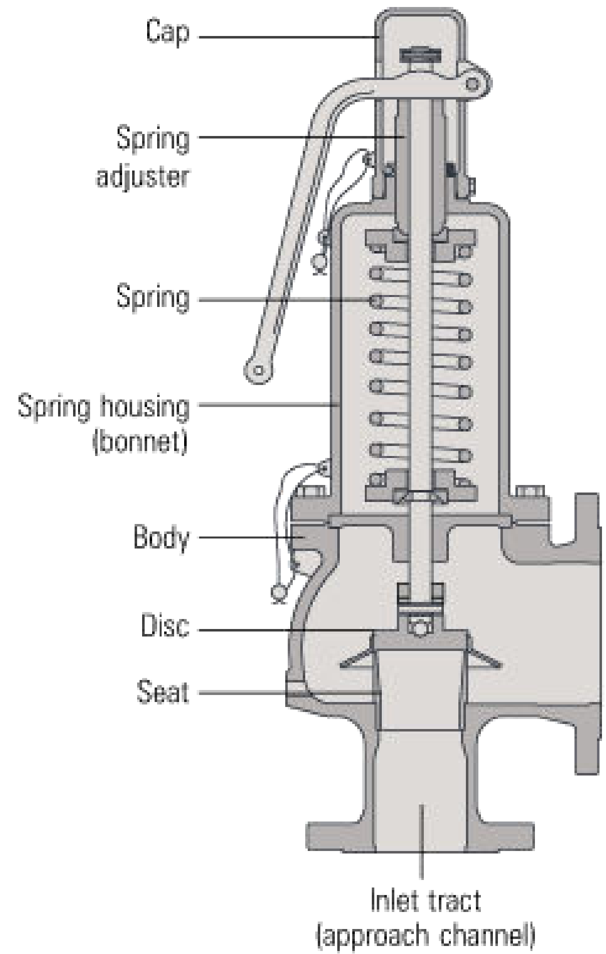
Typical Safety Valves



Full lift safety valve

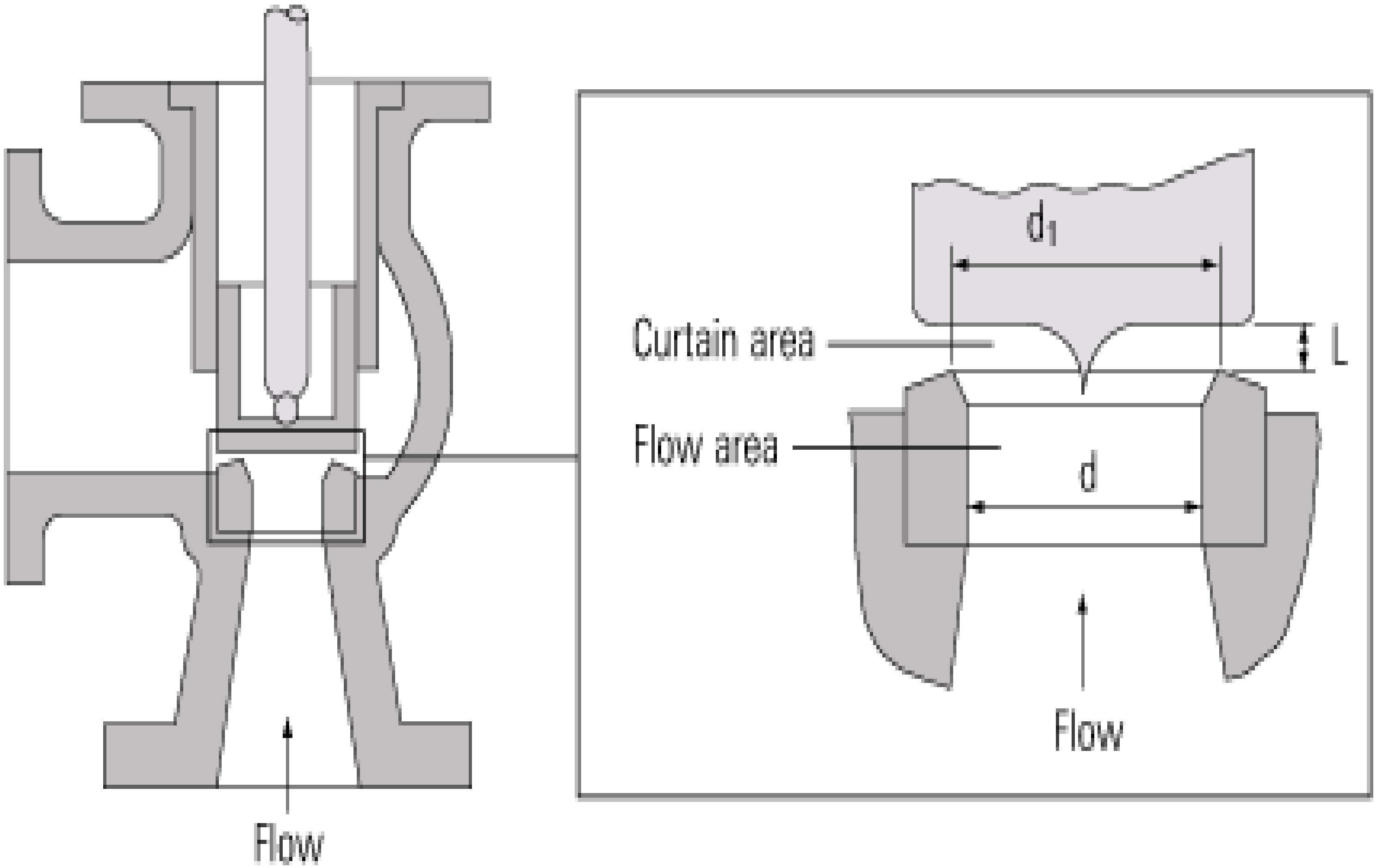


Typical ASME valve



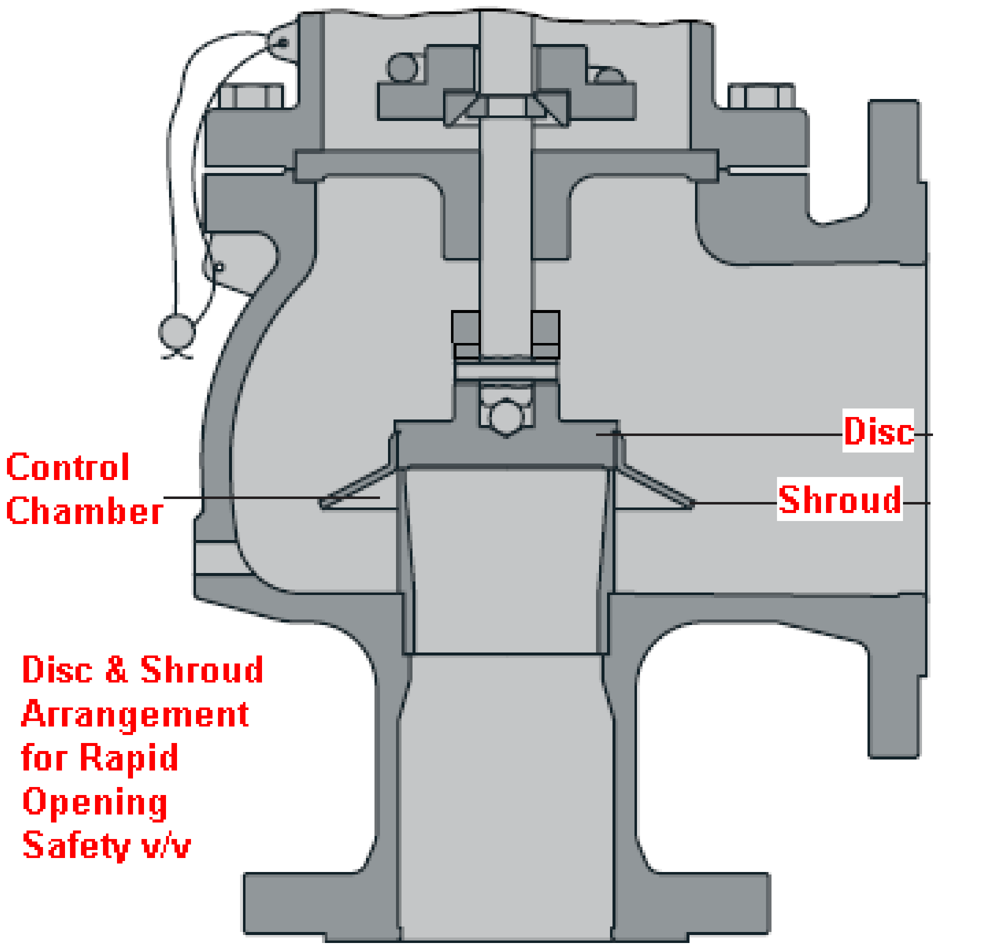
Typical DIN valve

Full Lift = L= D/4

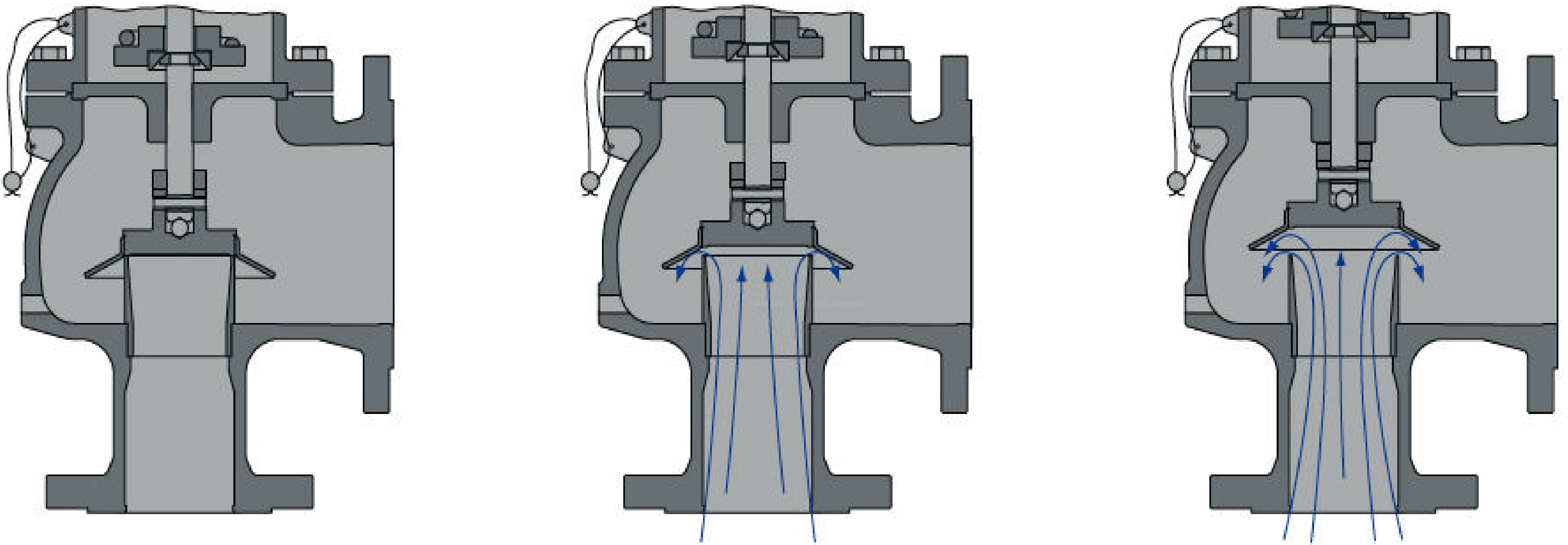


Safety Valve: Operation - Lifting

- In order to achieve full opening from this small overpressure, the disc arrangement has to be specially designed to provide rapid opening.
- This is usually done by placing a shroud, skirt or hood around the disc.
- The volume contained within this shroud is known as the control or huddling chamber.

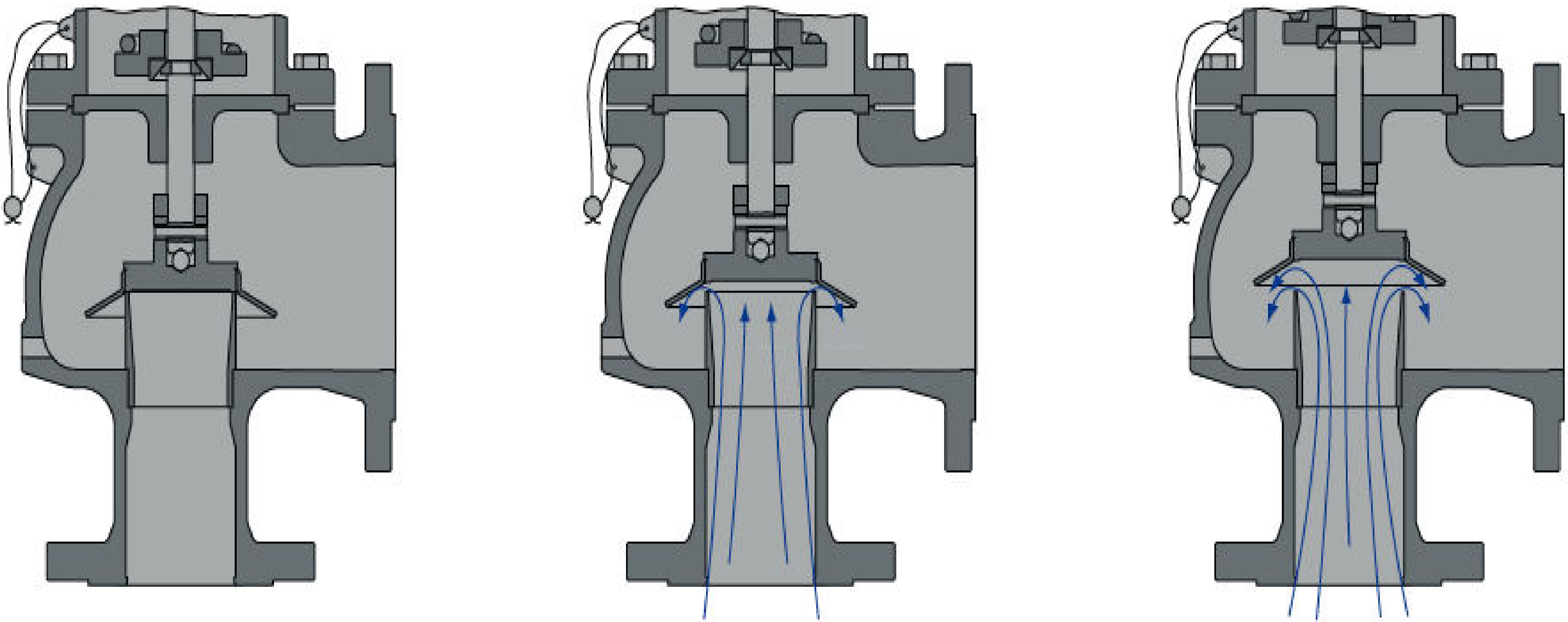


Safety Valve: Operation - Lifting



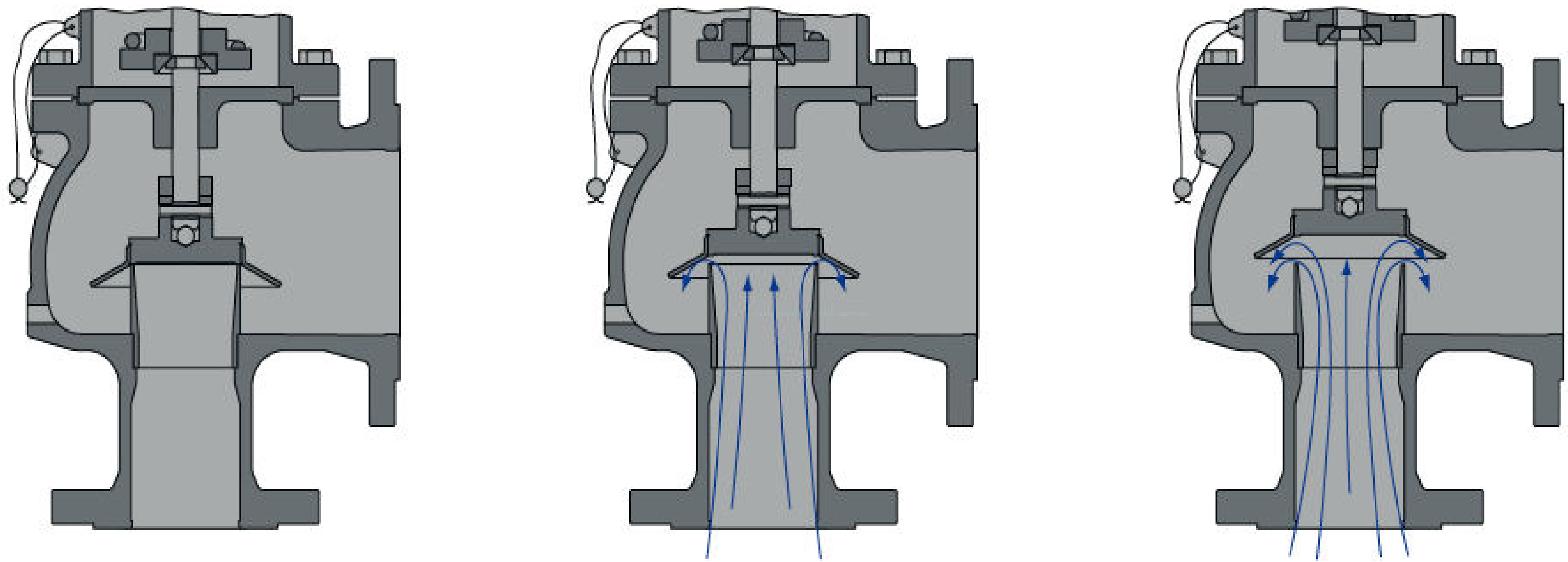
- As lift begins and fluid enters the chamber, a larger area of the shroud is exposed to the fluid pressure.

Safety Valve: Operation - Lifting



- Since the magnitude of the lifting force (F) is proportional to the product of the pressure (P) and the area exposed to the fluid (A); (F = P x A), the opening force is increased.

<u>Safety Valve: Operation - Lifting</u>



- This incremental increase in opening force overcompensates for the increase in spring force, causing rapid opening. At the same time, the shroud reverses the direction of the flow, which provides a reaction force, further enhancing the lift.

Safety Valve Operation – Re-seatting

- The difference between the set pressure and this reseating pressure is known as the ‘blowdown’
- It is usually specified as a percentage of the set pressure.
- For compressible fluids, the blowdown is usually less than 10%
- For liquids, it can be up to 20%.

REGULATION FOR SAFETY VALVE

- Number of safety valve.
- Size of safety valve.
- Superheater and economizer

Number of Safety Valves for Boiler

- Each boiler (including exhaust gas boiler) and steam generator is to be fitted with at least one safety valve and where the water-heating surface is more than 46.5 m^2 (500 ft^2), two or more safety valves are to be provided.
- The valves are to be of equal size as far as practicable and their aggregate relieving capacity is not to be less than the evaporating capacity of the boiler under maximum operating conditions.

Safety Valves – Size Restrictions

- In no case,
  - is the inlet diameter of any safety valve for propulsion boiler and superheaters used to generate steam for main propulsion and other machinery to be less than 38 mm (1.5 in.) nor more than 102 mm (4 in.).
  - For auxiliary boilers and exhaust gas economizers, the inlet diameter of the safety valve must not be less than 19 mm (3/4 in.) nor more than 102 mm (4 in.).

Safety Valves – for Superheater & Economizer

- *Superheater*. Each superheater, regardless of whether it can be isolated from the boiler or not, is to be fitted with at least one safety valve on the superheater outlet.
- *Economizers*. Each economizer, where fitted with a bypass, is to be provided with a sentinel relief valve, unless the bypass arrangement will prevent a buildup of pressure in the economizer when it is bypassed.

Pressure Setting – Boiler Drums

- At least one safety valve on the boiler drum is to be set at or below the maximum allowable working pressure.
- If more than one safety valve is installed, the highest setting among the safety valves is not to exceed the maximum allowable working pressure by more than 3%.
- The range of pressure settings of all the drum safety valves is not to exceed 10% of the highest pressure to which any safety valve is set.
- In no case is the relief pressure to be greater than the design pressure of the steam piping or that of the machinery connected to the boiler plus the pressure drop in the steam piping.

Pressure Setting – Superheaters

- Where a superheater is fitted, the superheater safety valve is to be set to relieve at a pressure no greater than the design pressure of the steam piping or the design pressure of the machinery connected to the superheater plus pressure drop in the steam piping.
- In no case is the superheater safety valve to be set at a pressure greater than the design pressure of the superheater.
- In connection with the superheater, the safety valves on the boiler drum are to be set at a pressure not less than the superheater-valve setting plus 0.34 bar (0.35 kgf/cm^2, 5 psi), plus approximately the normal-load pressure drop through the superheater.

Safety Valve – Easing Gear

- Each boiler and superheater safety valve is to be fitted with an efficient mechanical means by which the valve disc may be positively lifted from its seat.
- This mechanism is to be so arranged that the valves may be safely operated from the boiler room or machinery space platforms, either by hand or by any approved power arrangement.

Safety Valve – Connection to Boiler & Superheater

- Safety valves are to be connected directly to the boiler, except that they may be mounted on a common fitting.
- However, they are not to be mounted on the same fitting as that for the main or auxiliary steam outlet.
- This does not apply to super heater safety valves, which may be mounted on the fitting for the super heater steam outlet.

Safety Valve – Escape Pipe

- The area of the escape pipe is to be at least equal to the combined outlet area of all of the safety valves discharging into it.
- The pipe is to be so routed as to prevent the accumulation of condensate and is to be so supported that the body of the safety valve is not subjected to undue load or moment.

Safety Valve – Drain Pipe

- Safety valve chests are to be fitted with drain pipes leading to the bilges or a suitable tank.
- No valve or cock is to be fitted in the drain pipe.

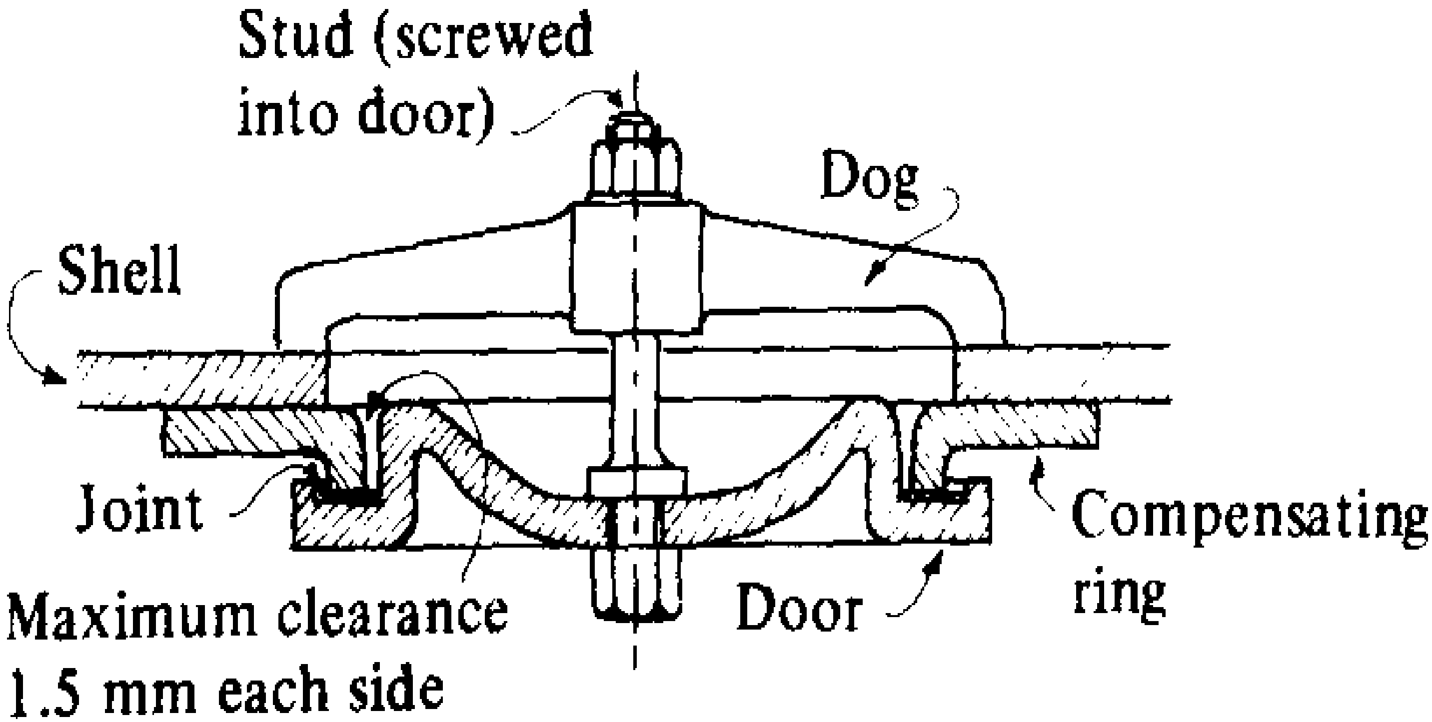
Safety Valve – Pressure Accumulation Test

- Safety valves are to be set under steam and tested with pressure accumulation tests in the presence of the Surveyor.
- The boiler pressure is not to rise more than 6% above the maximum allowable working pressure when the steam stop valve is closed under full firing condition for a duration of 15 minutes for firetube boilers and 7 minutes for watertube boilers.
- During this test, no more feed water is to be supplied than that necessary to maintain a safe working water level.
- The popping point of each safety valve is not to be more than 3% above its set pressure.

Safety Valve – Pressure Accumulation Test (waiver)

- Where such accumulation tests are impractical because of superheater design, an application to omit such tests may be approved, provided the following are complied with:
  - All safety valves are to be set in the presence of the Surveyor.
  - Capacity tests have been completed in the presence of the Surveyor on each valve
  - type.
  - The valve manufacturer supplies a certificate for each safety valve stating its capacity at the maximum allowable working pressure and temperature of the boiler.
  - The boiler manufacturer supplies a certificate stating the maximum evaporation of the boiler.
  - Due consideration is given to back pressure in the safety valve steam escape pipe.

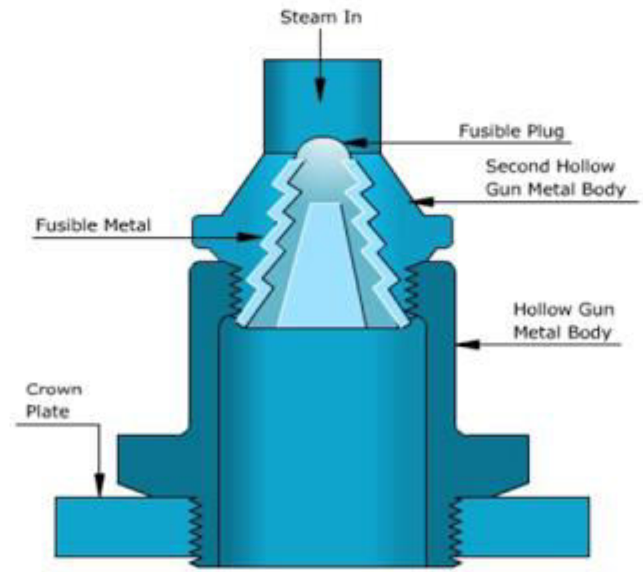
Manhole door



Fusible Plug

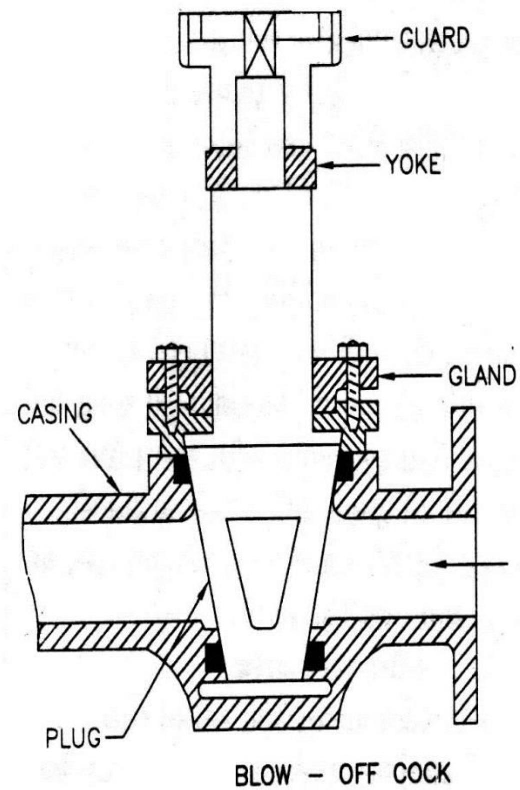
It is very important safety device, which protects the fire tube boiler against overheating. It is located just above the furnace in the boiler. It consists of gun metal plug fixed in a gun metal body with fusible molten metal.

During the normal boiler operation, the fusible plug is covered by water and its temperature does not rise to its melting state. But when the water level falls too low in the boiler, it uncovers the fusible plug. The furnace gases heat up the plug and fusible metal of plug melts, the inner plug falls down The water and steam then rush through the hole and extinguish the fire before any major damage occurs to the boiler due to overheating.



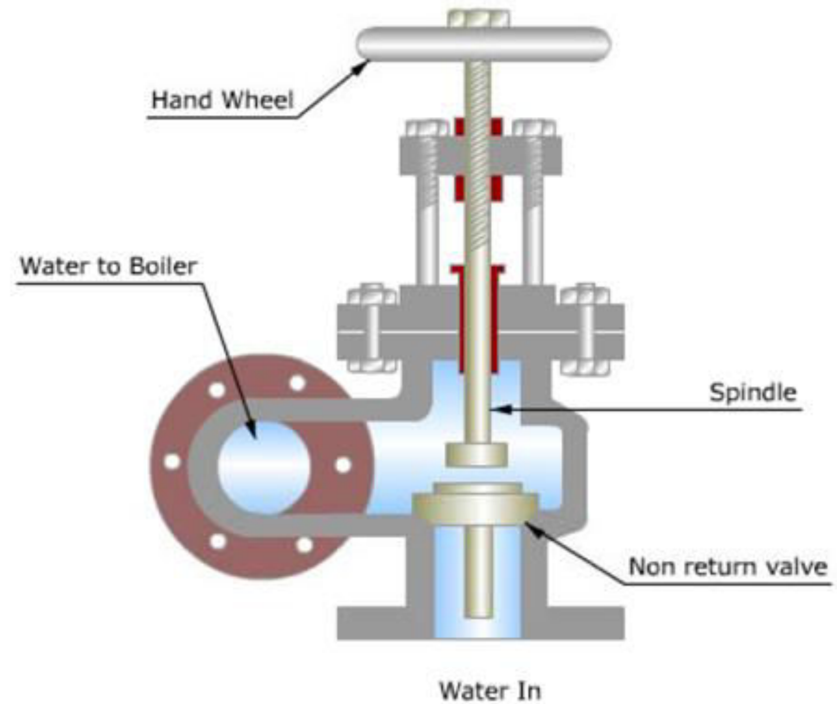
Blow-Off Cock

The function of blow-off cock is to discharge mud and other sediments deposited in the bottom most part of the water space in the boiler, while boiler is in operation. It can also be used to drain-off boiler water. Hence it is mounted at the lowest part of the boiler. When it is open, water under the pressure rushes out, thus carrying sediments and mud.



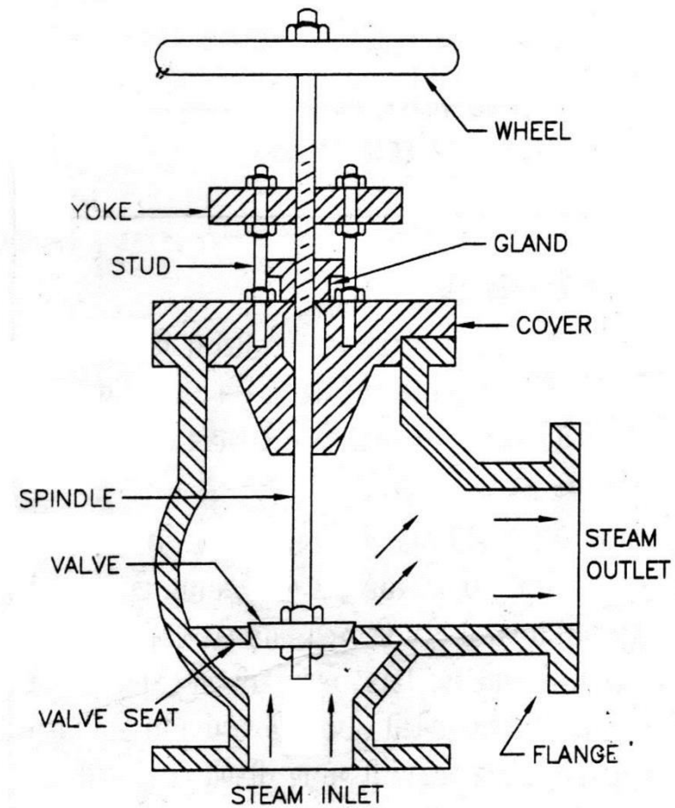
Feed Check Valve

The feed check valve is fitted to the boiler, slightly below the working level in the boiler. It
is used to supply high pressure feed water to boiler. It also prevents the returning of feed
water from the boiler if feed pump fails to work.



Steam Stop Valve

The steam stop valve is located on the highest part of the steam space. It regulates the steam supply to use. The steam stop valve can be operated manually or automatically.



STOP VALVE

Summary

- major mountings fitted on a boiler.
- significance of each of the mountings.
- Identify the various components of a gauge glass and safety valve.
- Describe the constructional features of various basic types of gauge glasses and safety valves.
- Explain the procedure for gauge glass blowing through.
- Explain the procedure for setting a safety valve’s set pressure and blowdown.
- Discuss the maintenance procedure for safety valve assembly.

Thank You

Module 4: Boilers and Internal Combustion Engine:
Boiler Mountings and Accessories, Fire Tube and Water Tube Boilers, Cochran Boiler, Babcock and Wilcox Boiler.

Model questions on Boilers

9.1. State how the boilers are classified.

9.2. Explain the principle of fire tube and water tube boilers.

9.3. Describe with a neat sketch the working of Cochran boiler. Show the position of different mountings and explain the function of each.

9.4. Describe, giving neat sketches, the construction and working of a Lancashire boiler. Show the positions of different mountings and accessories.

9.5. Sketch and describe the working of a Locomotive boiler. Show the positions of fusible plug, blow off cock, feed check valve and superheater. Mention the function of each. Describe the method of obtaining draught in this boiler.

9.6. Give an outline sketch showing the arrangement of water tubes and furnace of a Babcock and Wilcox boiler. Indicate on it, the path of the flue gases and water circulation. Show the positions of fusible plug, blow off cock and superheater. Mention the function of each.

9.7. Explain why the superheater tubes are flooded with water at the starting of the boilers ?

9.8. Mention the chief advantages and disadvantages of fire tube boilers over water tube boilers.

9.9. Discuss the chief advantages of water tube boilers over fire tube boilers.

9.10. What are the considerations which would guide you in selecting the type of boiler to be adopted for a specific purpose ?

9.11. Distinguish between water-tube and fire-tube boilers and state under what circumstances each type would be desirable.

Model questions on Boiler Mountings and Accessories

10.1. Describe with a neat sketch water level indicator and explain how the flow of steam and water is automatically stopped when the glass tube breaks.

10.2. Explain why safety valves are needed in a boiler. Draw a neat sketch of spring-loaded safety valve and explain its working.

10.3. What do you understand by high steam, low water safety valve ? Explain its working with a neat sketch.

10.4. What are the major differences between mountings and accessories ? Give three examples of each.

10.5. Draw a neat sketch of a Bourdon pressure gauge and explain its working principle.

10.6. What is the purpose of steam stop valve ? Explain its working.

10.7. What is the purpose of a fusible plug ? Explain its working. Indicate its usual location for Cochran, Lancashire, Locomotive and Babcock and Wilcox boilers.

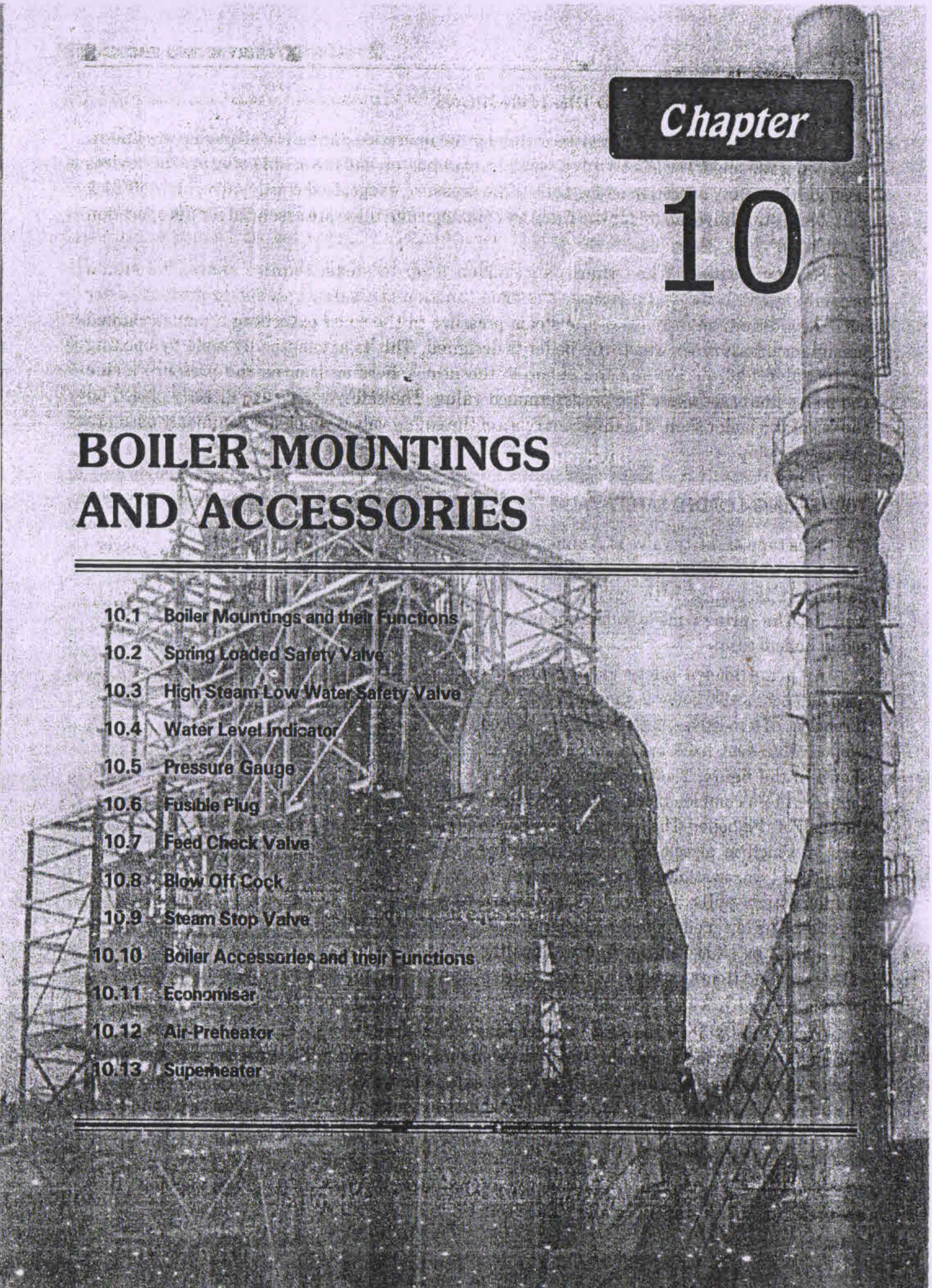
10.8. Explain why the blow-off cock is operated periodically when the boiler is working. Where is it located ? Explain its working with a neat sketch.

10.9. What do you understand by feed check valve ? Explain the working of a feed check valve with a neat sketch.

10.10. Draw a neat diagram of Green Economiser and explain its working.

10.11. What are the advantages of preheating the air ? Draw a neat diagram of tubular air-preheater and explain its working.

10.12. Explain the functions of a superheater. With a neat sketch, describe the working of a superheater used in Lancashire Boilers.



Chapter

10

BOILER MOUNTINGS AND ACCESSORIES

10.1 BOILER MOUNTINGS AND THEIR FUNCTIONS

Different fittings and devices necessary for the operation and safety of a boiler are known as boiler mountings. The safety valve, water level indicator, and the fusible plug are the devices used for the safety operation of the boiler. The pressure gauge, feed check valve, blow-off cock and steam stop valve fall under the category of fittings and these are essential for the operation of the boiler.

Safety Valves. When there is a sudden drop in steam requirements, the steam pressure in the boiler will increase. The main function of a safety valve is to prevent under such a condition, an increase in the steam pressure in the boiler exceeding a predetermined maximum pressure for which the boiler is designed. This is automatically done by opening of the valve and discharging the steam to the atmosphere as soon as the pressure inside the boiler increases above the predetermined value. The safety valves are directly placed on the top of the boiler shell. The different types of the safety valves which are commonly used are discussed below.

10.2 SPRING-LOADED SAFETY VALVE

This type of safety valve is commonly used now-a-days for stationary as well as mobile boilers. It is loaded with spring instead of weights. The spring is made from a square steel rod in helical form.

A spring loaded safety valve commonly used on Locomotive boiler is shown in Fig. 10.1. It consists of two valves, each of which is placed over a valve seat fixed over a branch pipe as shown in the figure. The two branch pipes are connected to a common block which is fixed on the shell of the boiler. The lever has two pivots each of which is placed over each respective valve. The lever is attached with a spring at its middle which pulls the lever in downward direction. The lower end of the spring is attached to the back as shown in the figure. Thus the valves are held tight to their seats by the spring force.

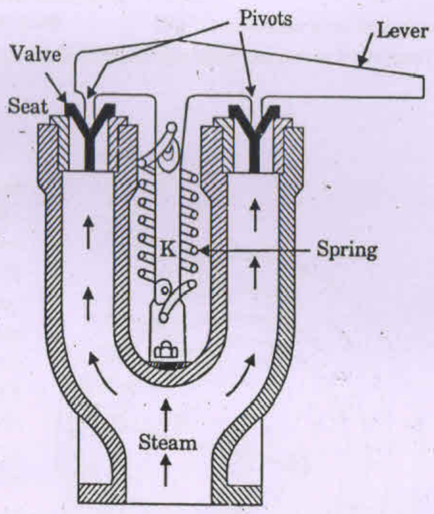


Fig. 10.1. Spring loaded safety valve.

These valves are lifted against the spring when the steam pressure is greater than the working pressure and allows the steam to escape from the boiler till the pressure in the boiler reaches its working pressure. The lever has an extension which projects into the driver's cabin. The driver can release the pressure if required just by raising the lever. The lever is connected loosely by a link to the block. This limits the valve opening and prevents the lever blowing off in case of spring failure.

The valve is much lighter and compact compared with other safety valves; therefore, they are preferred on all stationary and mobile boilers.

10.3 HIGH STEAM LOW WATER SAFETY VALVE

It is a combined safety arrangement against high steam pressure and low water level in the boiler. This is generally used in Lancashire boiler. It is actually a combination of two valves, one of which is lever safety valve which blows off steam when the working pressure of steam exceeds the preset value. The second valve operates by blowing off the steam when the water level in the boiler falls below predetermined level. This raises an alarm, so that corrective action can be taken by pumping the water in the boiler.

A commonly used high steam low water safety valve is shown in Fig. 10.2. It consists of two valves V_1 and V_2 of which the valve V_1 rests upon the valve seat and valve V_2 is placed over the valve V_1 which acts as valve seat for valve V_2.

To maintain the pressure of the steam in the boiler, it acts like a simple lever safety valve. The lever L_1 is hinged at its one end and a weight W is hung from the other end as shown in the figure. The pivot P is placed over a valve V_1 and when the steam pressure exceeds the normal limit, the valve V_1 is lifted with the lever L_1 and the valve V_2 also. So the steam escapes through the passage between the value seat and the valve V_1 till the pressure becomes normal.

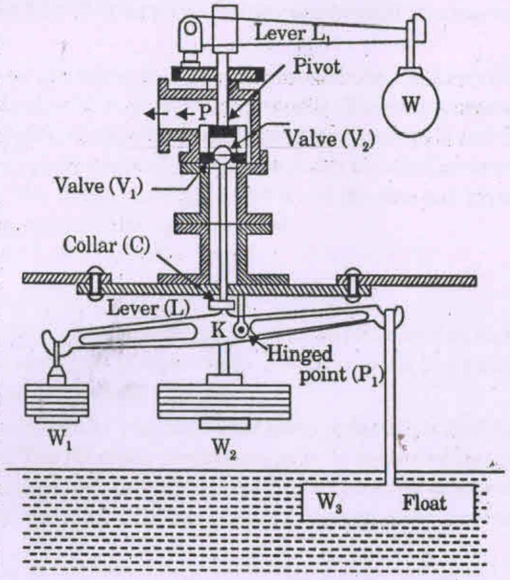


Fig. 10.2. High steam and low water safety valve.

Inside a boiler shell a lever L is hinged at the point P_1 as shown in figure. A weight W_1 is connected to one end of the lever and a large weight $W_3 > W_1$ to the other end. The valve V_2 is connected with a spindle, the lower end of which has a weight W_2. The knife edge K touches a collar C of the spindle of the valve V_2. Under the normal water level, the weight W_3 remains in water and whole system with weights is balanced. When the level of the water falls and the weight W_3 moves in a downward direction (because its acting weight increases as buoyancy force is decreased as it is partly uncovered). This causes the knife edge to push the collar C with the valve V_2 in the upward direction. So the steam escapes through the passage between the valves V_1 and V_2. The escaping steam causes a loud noise as it passes through a narrow passage. This noise warns the boiler attendant about the low water level of the boiler, so that corrective action may be initiated.

10.4 WATER LEVEL INDICATOR

It is an important fitting which indicates water level inside the boiler to the observer. Usually two water level indicators are fitted in front of the boiler. The water indicator shows the level or water in the boiler drum and warns the operator if by chance the water level goes below a fixed mark, so that corrective action may be taken in time to avoid any accident.

A water level indicator used in low pressure boilers is shown is Fig. 10.3. It consists of three cocks and a glass tube. The steam cock 1 keeps the glass tube in connection with the steam space and cock 2 puts the glass tube in connection with the water space in the boiler. The drain cock '3' is used to drain out the water from the glass tube at intervals to ascertain that the steam and water cocks are clear in operation. The glass tube is generally protected with a shield.

For the observation of the water level in the boiler, the steam and water cocks are opened and drain cock is closed. In this case, the handles are placed in vertical position as shown in Fig. 10.3. The steam enters from the upper end of the glass tube and water enters from the lower end of the tube. So the water level inside the boiler will be same as seen in the glass tube.

The rectangular passage at the ends of the glass tube contains two balls. In case, the glass tube is broken, the balls are carried along its passage to the ends of glass tube and flow of water and steam out of the boiler is prevented.

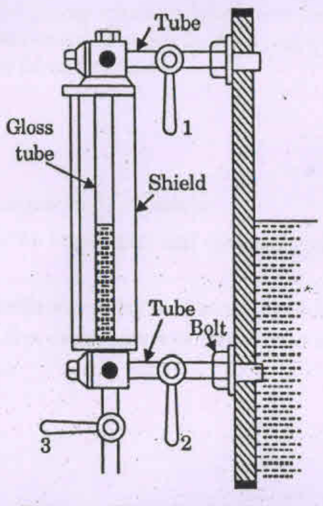


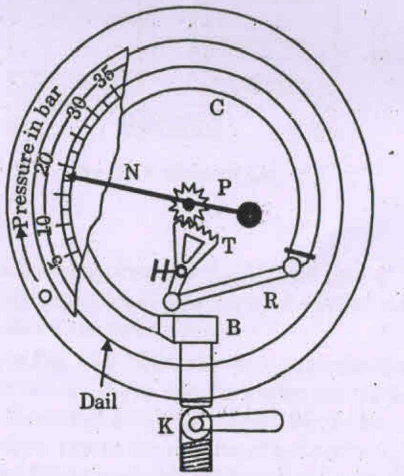
Fig. 10.3. Water level indicator.

The water level indicator can be taken out from boiler for cleaning purposes by removing the bolts when the boiler is not working.

10.5 PRESSURE GAUGE

A pressure gauge is used to measure the pressure of the steam inside the boiler. The commonly used pressure gauge known as Bourdon type pressure gauge, is shown in Fig. 10.4. It consists of an elastic metallic type of elliptical cross-section and is bent in the form of circular arc. One end of the tube is fixed and connected to the steam space of the boiler and other end is connected to a sector wheel through a link. The sector remains in mesh with a pinion fixed on a spindle. A pointer as shown in the figure is attached to the spindle to read the pressure on a dial gauge.

When high pressure steam enters the elliptical tube, the tube section tries to become circular which causes the other end of the tube to move outward. The movement of the closed end of the tube (other end) is transmitted and magnified by the link and sector as shown in the figure. The sector is hinged at the point H as shown in the figure. The magnitude of the movement is indicated by the pointer on the dial.



C—Spring tube, B—Block, R—Rod, H—Hinged point, T—Toothed sector, P—Pinion, N—Pointer, K-Cock.

Fig. 10.4. Bourdon pressure gauge.

10.6 FUSIBLE PLUG

The main object of the fusible plug is to put off the fire in the furnace of the boiler when the water level in the boiler falls below an unsafe level and thus avoids the explosion which may take place due to overheating of the tubes and shell. This plug is generally fitted over the crown of the furnace or over the combustion chamber.

A fusible plug which is commonly used is shown in Fig. 10.5. A is a hollow gun metal body screwed into the crown of the boiler grate. B is a second hollow gunmetal plug screwed into the plug A. The third plug C is locked with plug B by pouring a low melting point metal into groove provided for the same.

Under normal water level condition in the boiler, this plug is covered with water which keeps the temperature of the fusible metal below its melting point. But when the water level in the boiler falls low enough to uncover the plug; the fusible metal between the plugs B and C quickly melts and the plug C drops out. The opening so made allows the steam to rush the water into the furnace and extinguish the fire. The steam rushing out puts out the fire and gives warning that the crown of the furnace is in danger of being overheated.

10.7 FEED CHECK VALVE

The function of the feed check valve is to allow the supply of water to the boiler at high pressure continuously and to prevent the back flow of water from the boiler when the pump pressure is less than boiler pressure or when pump fails.

A commonly used feed check valve is shown in Fig. 10.6. It is fitted to the shell slightly below the normal water level of the boiler. The lift of the non-return valve is regulated by the end position of the spindle (E) which is attached with the hand wheel. The spindle can be moved upward or downward with the help of hand wheel as the upper portion of the spindle is screwed to a nut.

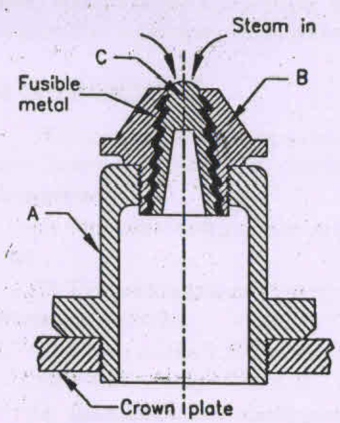


Fig. 10.5. Fusible plug.

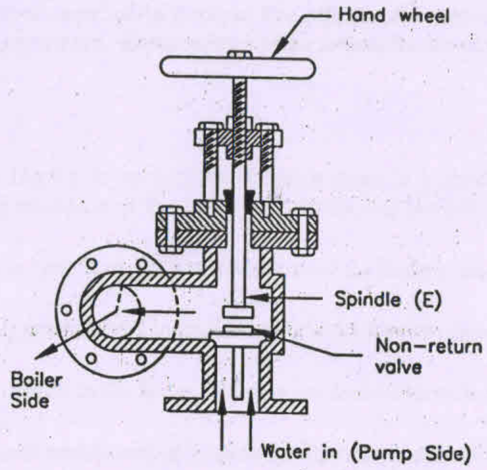


Fig. 10.6. Feed check valve.

At normal working condition, the non-return valve is lifted due to the pressure of water from the pump and the water is fed to boiler. But when the pump pressure falls below boiler pressure or if the pump stops, non-return valve is closed automatically due to the pressure of the steam from the boiler and prevents the escape of water from the boiler.

10.8 BLOW-OFF COCK

The blow-off cock is used for dual functions :

1. To empty the boiler when necessary for cleaning, repair and inspection.

2. To discharge the mud and sediments carried with the feed water and accumulated at the bottom of the boiler.

By periodic blow-off, the salt concentration in the boiler is also reduced. Even with a small amount of dissolved salt, over a period of time, due to the evaporation of water, the salt accumulates in the boiler, raising the salt concentration.

It is fitted to the lowest part of the boiler either directly with the boiler shell or to a pipe connected with the boiler.

A commonly used type of blow-off cock is shown in Fig. 10.7. It consists of a conical plug fitted accurately into a similar casing. The plug has a rectangular opening which may be brought with the line of the passage of the casing by rotating the plug. This causes the water to be discharged from the boiler. The discharging of water may be stopped by rotating the plug again.

The blow-off cock should be operated only when the boiler is on if the sediments are to be removed. This is because, the sediments are forced out quickly due to the high steam pressure in the boiler.

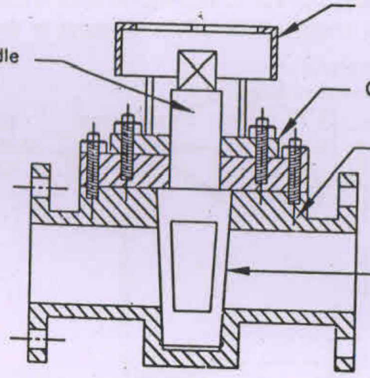


Fig. 10.7. Blow-off cock.

10.9 STEAM STOP VALVE

It is the largest valve on the steam boiler and usually fitted to the highest part of the boiler shell. The function of the stop valve is to regulate the flow of steam from the boiler to the engine as per requirement and shut off the steam flow when not required.

A commonly used steam stop valve is shown in Fig. 10.8. The main body is made of cast steel. The valve, valve seat and the nut through which the valve spindle works, are made of brass for smooth working. The spindle is passed through a gland to prevent the leakage of steam. The spindle is rotated by means of a hand wheel. Due to the rotation of hand-wheel, the valve may move up or down and it may close or open the passage fully or partially for the flow of the steam. In locomotive boiler, the flow of steam is controlled by means of a regulator (as discussed in previous chapter) which is placed inside the boiler shell and operated by a handle from the driver's cabin.

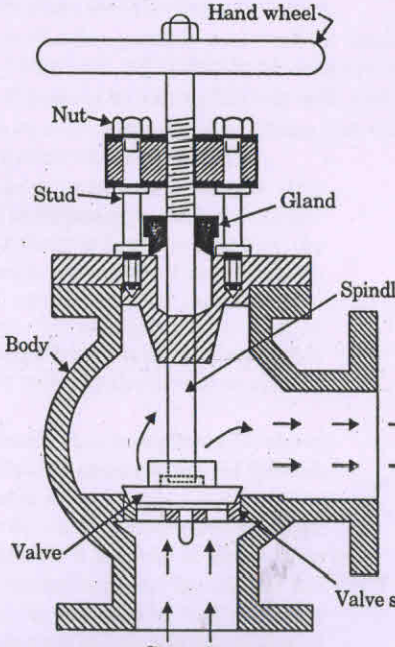


Fig. 10.8. Steam stop valve.

10.10 BOILER ACCESSORIES AND THEIR FUNCTIONS

Boiler accessories are auxiliary parts required to increase the overall efficiency of the boiler. Water feeding equipments, air-preheaters, economisers and superheaters are some of the essential accessories of the boiler.

10.11 ECONOMISER

The economiser usefully extracts the waste heat of the chimney gases to preheat the water before it is fed into the boiler. Preheating of the boiler feed water has the following advantages :

(1) There is a saving of fuel as waste heat from the flue gases is used for heating the feed water.

(2) Dissolved gases as air and CO_2 are removed by preheating the feed water, reducing corrosion and pitting.

(3) There will be less temperature strain in the boiler plates as the feed water enters the boiler at a higher temperature.

(4) Circulation of the water is very well maintained as quick evaporation is possible because of hot feed water.

(5) This unit improves the overall efficiency of the boiler by reducing the fuel consumption.

A Green's economiser commonly used in Lancashire boiler is shown in Fig. 10.9(*a*). The economiser consists of vertical cast iron pipes which are fitted with two headers, one at the bottom and other at the top. The feed water is passed through the bottom header, economiser pipes and top header and on to the boiler. The hot gases pass over the external surface of the water tubes. The heat from the hot gases is given to the feed water through the tube surface. A safety valve is fitted on the top header for the safety of pipes against any highest pressure of water that may be developed. A blow-off valve (not shown in the figure) is also fitted at the lowest point of the economiser to discharge the sediments collected from the feed water.

To prevent the deposition of soot from the flue gases over the economiser tubes, a set of scrapers is fitted over the pipes as shown in the figure. This is necessary as the deposition of soot reduces drastically the heat flow rate from the gases to the water. The soot is removed by moving the scrapers over the pipes up and down continuously with the help of chain and gear arrangement. The soot removed from the pipes is collected in soot chamber situated below the bottom header and removed periodically.

The temperature of the feed water should not be less than 35°C because there is a danger of corrosion of the cold pipe outer surfaces due to the condensation of moisture and SO_2 contained in the flue gases.

By-pass arrangement for the furnace gases must always be provided so that the economiser may be put out of action when necessary (for repair or inspection). Fig. 10.9(*b*) shows such an arrangement used with Lancashire boiler.

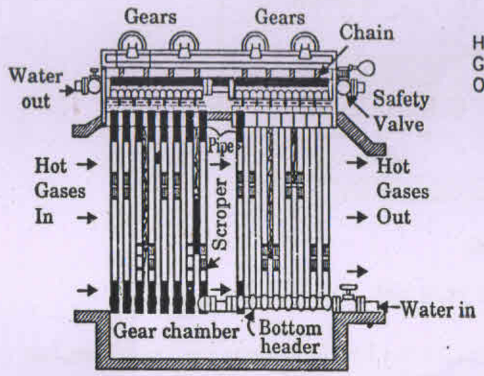


Fig. 10.9(*a*). Green Economiser.

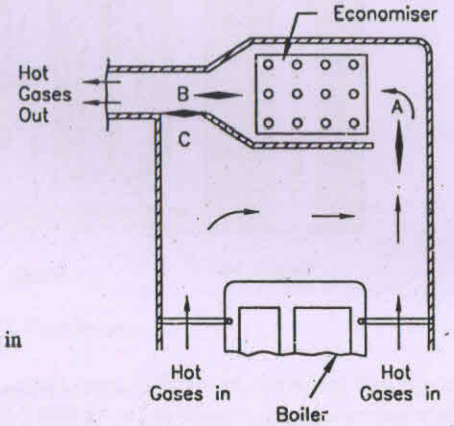


Fig. 10.9(*b*). Bypass arrangement.

(*a*) When hot gases are used for heating the feed water, the dampers *A* and *B* are kept open and damper *C* is closed.

(*b*) When hot gases are to be bypassed for economiser inspection, the dampers *A* and *B* are closed and damper *C* is kept open.

10.12 AIR PREHEATER

Air preheater, like economiser, recovers some portion of the waste heat of the flue gases. Air supplied to the combustion chamber is preheated by using the heat in the waste flue gases. Air preheaters are placed after the economiser and before the gases enter the chimney.

The preheating of air offers the following advantages :

(1) Waste heat from the flue gases is recovered for heating air and reduces the fuel consumption by about 1.5% for each 100°C drop in gas temperature.

(2) Inferior grades of coal can be burnt efficiently with preheated air.

(3) Combustion can be more efficient and an intense flame can be achieved in the furnace. This increases the evaporation rate of the boiler.

Air-preheaters can be classified as tubular type, plate type and generative type. In tubular type, the hot flue gases pass through the tubes and air is forced to flow over the tubes. The plate type consists of a number of parallel plates. Air flows through the alternate spaces of the plates and flue gas passes through the remaining passages. In regenerative type, a wire mesh rotor is alternately heated by the flue gases and then cooled by the air and as a result, the air is heated.

A tubular type commonly used in smaller boiler plants is shown in Fig. 10.10. The hot gases are passed through the tubes and air circulates around them. Air is forced to deflect by using baffles as shown in the figure and compelled to move in a zigzag path for a number of times. This increases the period of contact between the air and hot surface and air is effectively heated. The soot and other material carried with gases are collected in the hopper at the bottom and removed periodically through the soot gate.

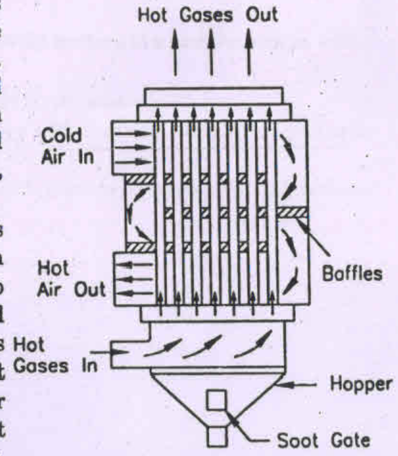


Fig. 10.10. Air preheater.

10.13 SUPERHEATER

Superheaters are used in boilers to increase the temperature of the steam above its saturation temperature. This is done by passing the steam through a small set of tubes and hot gases over them. Superheated steam is absolutely essential for power generation.

The advantages of superheated steam are listed below :

(1) It reduces the specific steam consumption of engines and turbines.

(2) It reduces the condensation losses in the pipes and engine cylinder.

(3) It eliminates the erosion of the turbine blades in the last stages.

(4) The efficiency of the steam plant is higher with the use of superheated steam.

The superheaters used in locomotive boilers and Babcox and Wilcox boilers have already been described.

The superheater commonly used in Lancashire boiler is shown in Fig. 10.11. This superheater consists of two headers and a set of superheater tubes made of high quality steel in the form of U-tube. Superheater is located in the path of furnace gases where the temperature of the gases is not less than 500°C. The superheater is located just before the gases enter the bottom flue.

The amount of hot gases passed over the superheater tubes should be in proportion to the amount of superheated steam passing through the tubes. Otherwise, the tubes would be overheated. To avoid this, the amounts of hot gases are diverted as shown in the figure. The superheater is put out of action by turning the damper upward to the vertical position. In this position of the damper, the gases coming out from the central flue pass directly into the bottom flue without passing over the superheater tubes.

The arrangement for getting superheated or wet steam is shown in the figure. For getting superheated steam, the valves A and B are opened and valve C is closed. And the damper is kept open as per the quantity of steam flowing through the pipe. For this position, the flow direction of the steam is shown is the figure. The steam from boiler enters into superheater through the valve A nad comes out through the valve B. If wet steam in required, then the valves A, B

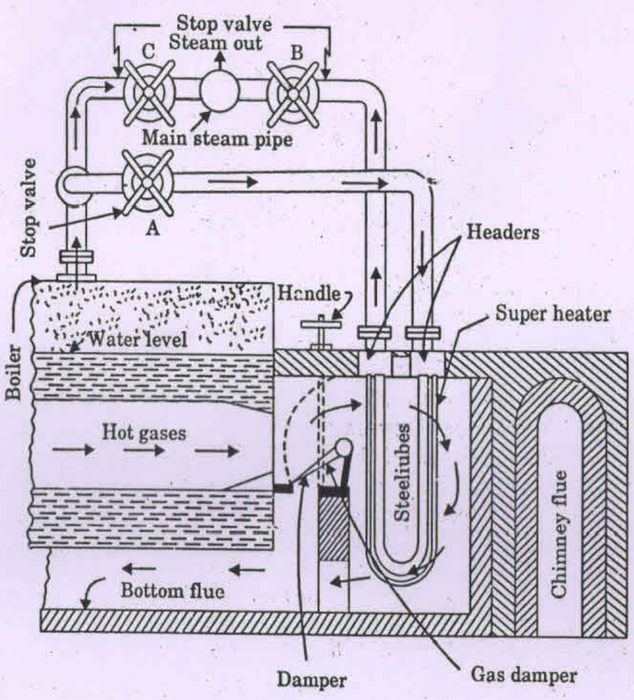


Fig. 10.11. Superheater.

and gas damper are kept closed, and valve C is kept open. In this case, the steam directly comes out from the boiler through the valve C. By adjusting the gas damper, the temperature of the steam coming out of superheater is always maintained constant irrespective of amount of steam passing through the superheater.

EXERCISES

10.1. Describe with a neat sketch water level indicator and explain how the flow of steam and water is automatically stopped when the glass tube breaks.

10.2. Explain why safety valves are needed in a boiler. Draw a neat sketch of spring-loaded safety valve and explain its working.

10.3. What do you understand by high steam, low water safety valve ? Explain its working with a neat sketch.

10.4. What are the major differences between mountings and accessories ? Give three examples of each.

10.5. Draw a neat sketch of a Bourdon pressure gauge and explain its working principle.

10.6. What is the purpose of steam stop valve ? Explain its working.

10.7. What is the purpose of a fusible plug ? Explain its working. Indicate its usual location for Cochran, Lancashire, Locomotive and Babcock and Wilcox boilers.

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10.10. Draw a neat diagram of Green Economiser and explain its working.

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10.12. Explain the functions of a superheater. With a neat sketch, describe the working of a superheater used in Lancashire Boilers.